Electronics and digital logic

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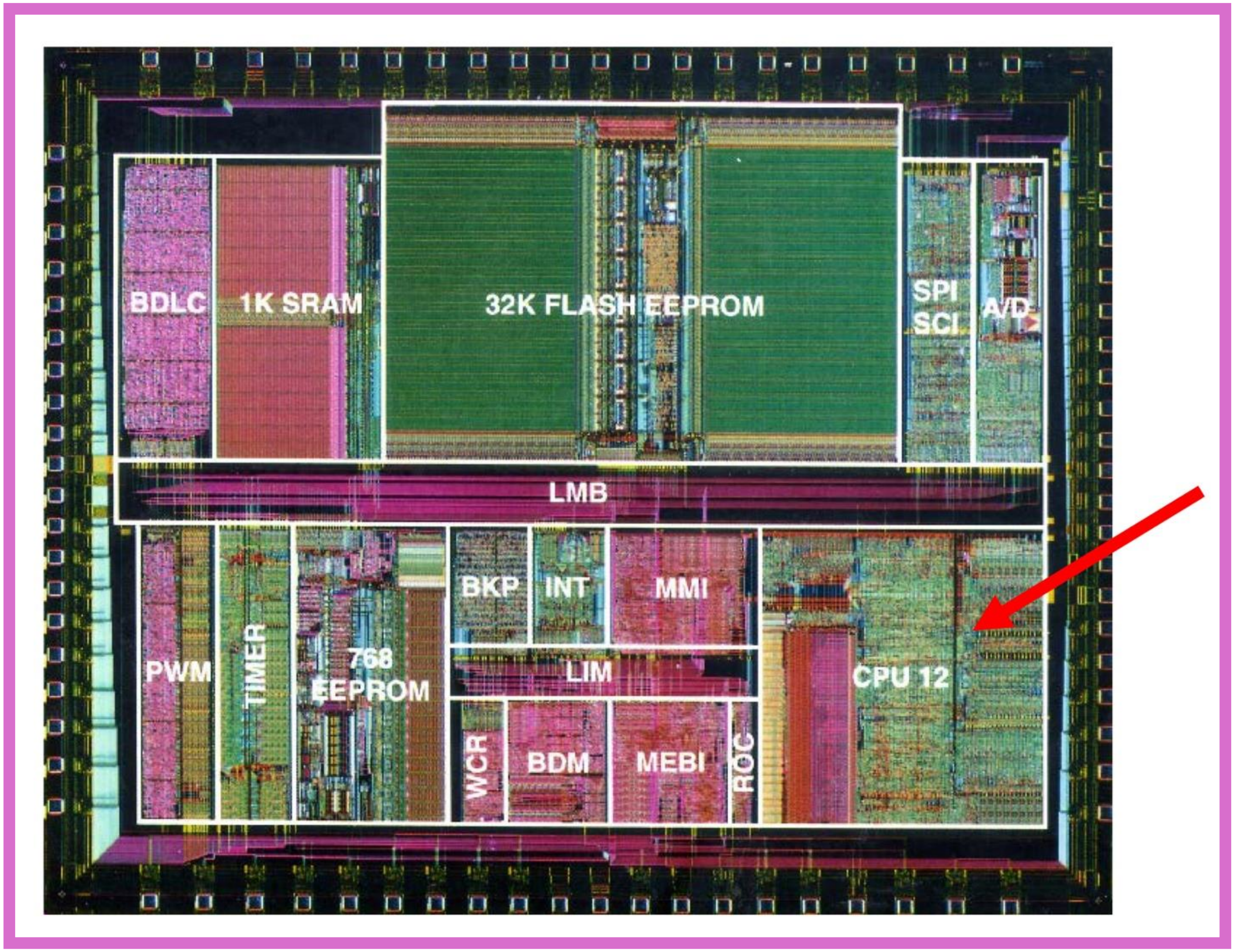
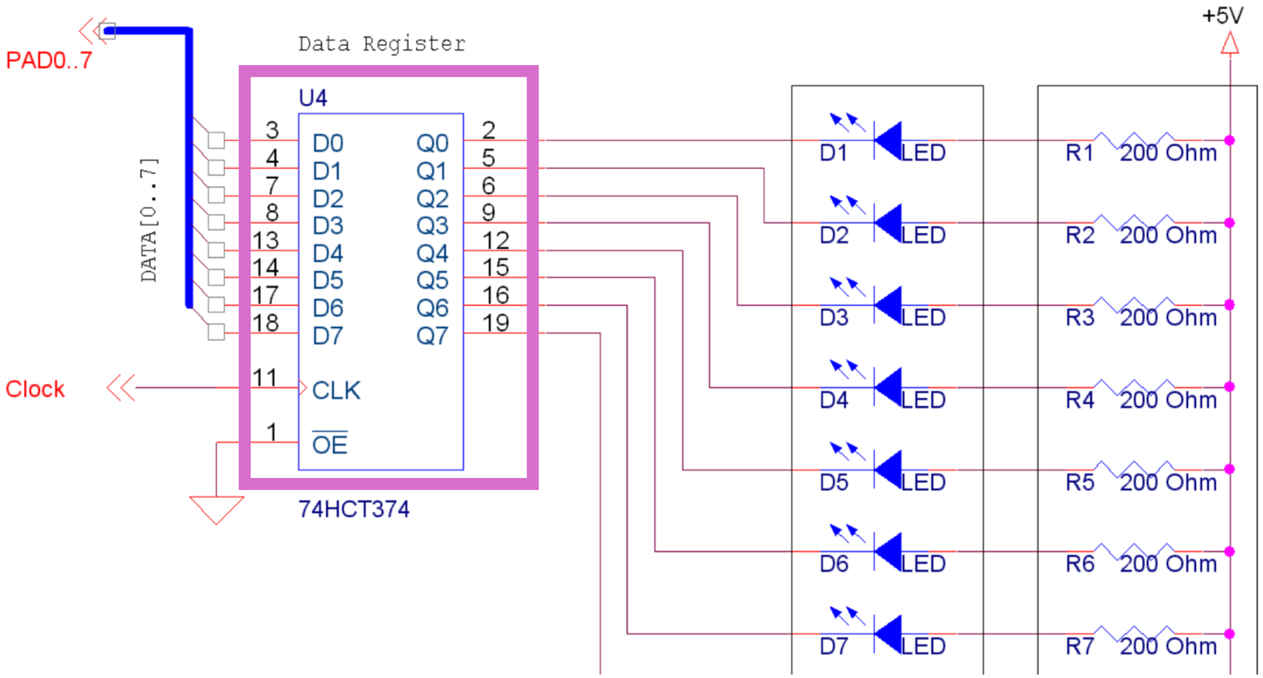
Recap: Microcontroller chip in practice

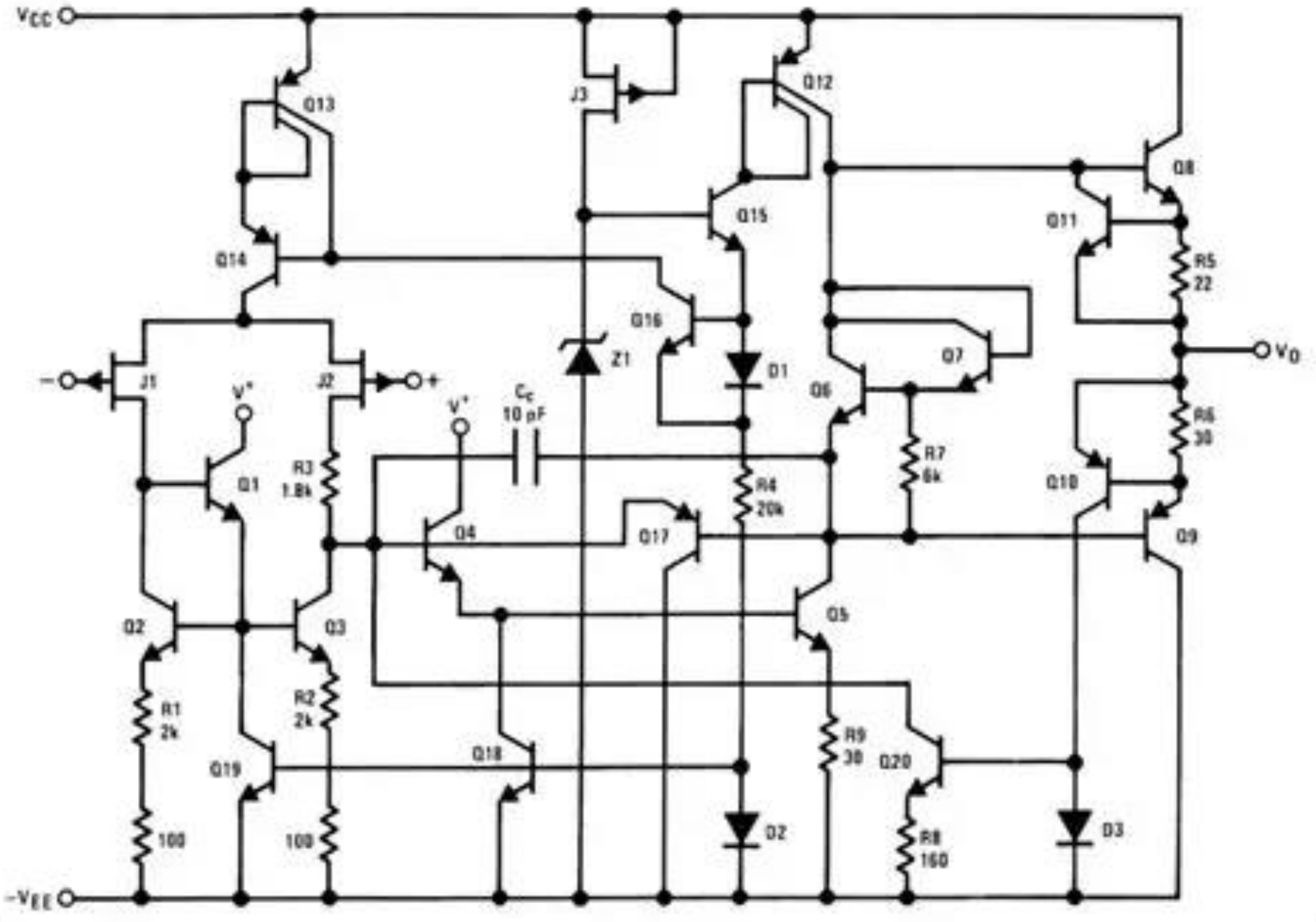


Chip diagram

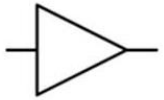


Fundamental unit: Electronic circuit (transistor)

Circuit diagram

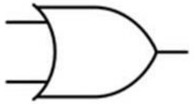


Digital logic

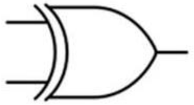


Input	Output
0	0
1	1

A	B	Output
0	0	0
1	0	0
0	1	0
1	1	1

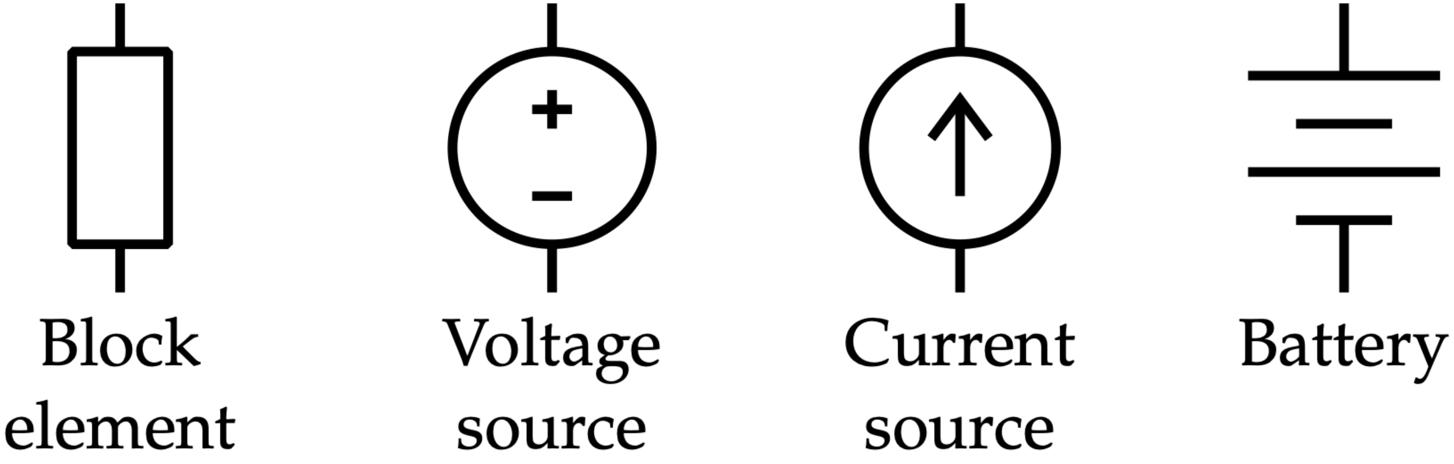


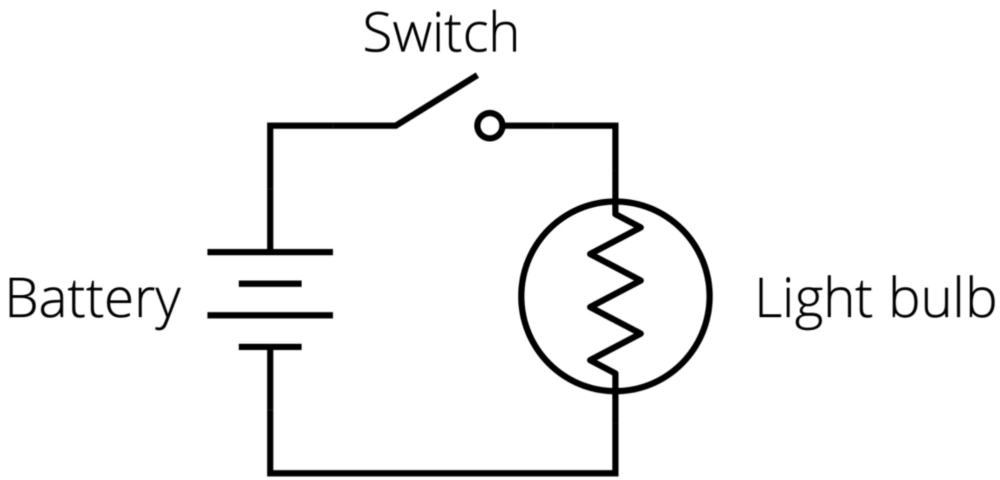
A	B	Output
0	0	0
1	0	1
0	1	1
1	1	1

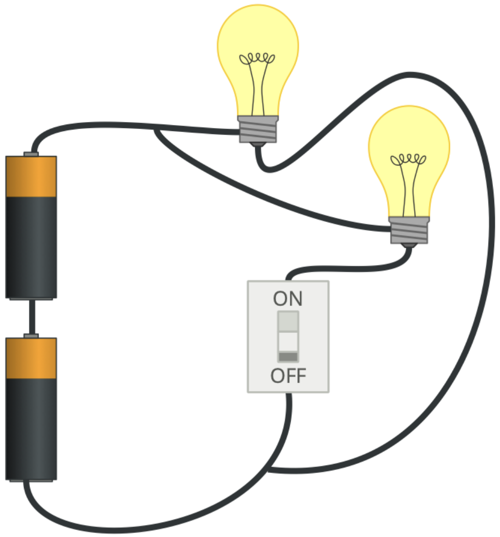


A	B	Output
0	0	0
1	0	1
0	1	1
1	1	0

Circuit schematics

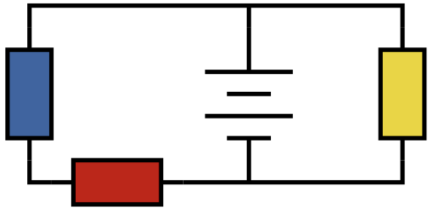






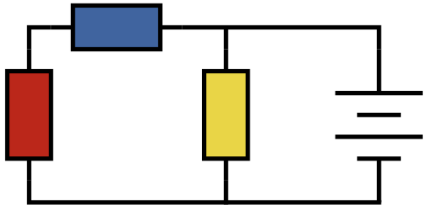
Slide acknowledgement:
Steven Bell, Chuan-Zheng Lee @ Stanford

Congruent schematics

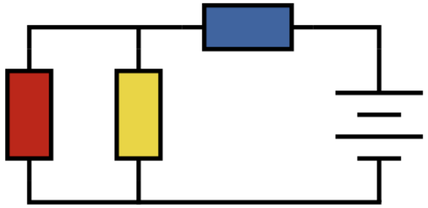


Which of the circuits below are congruent with the circuit above?

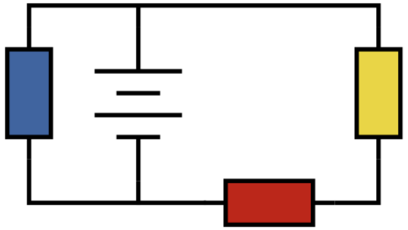
A



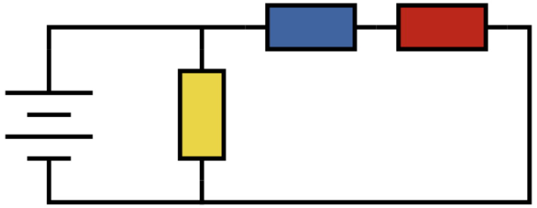
B



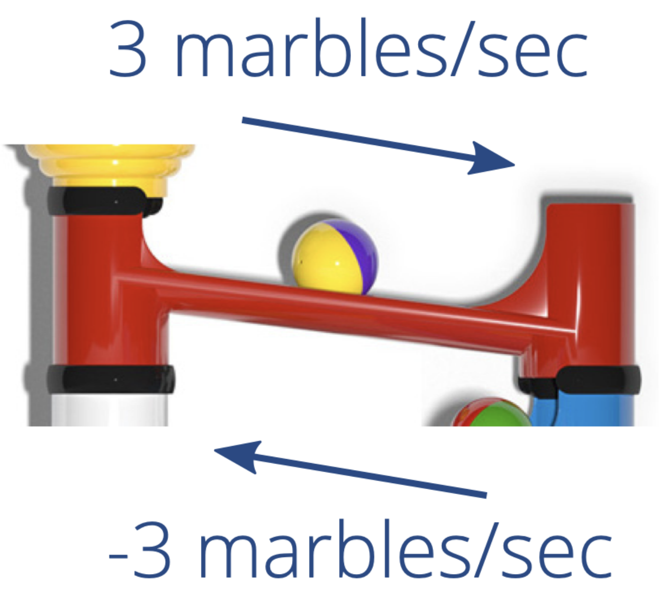
C

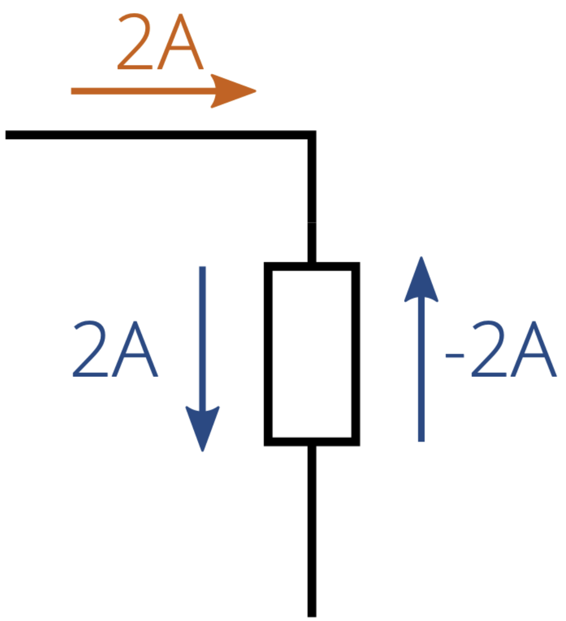


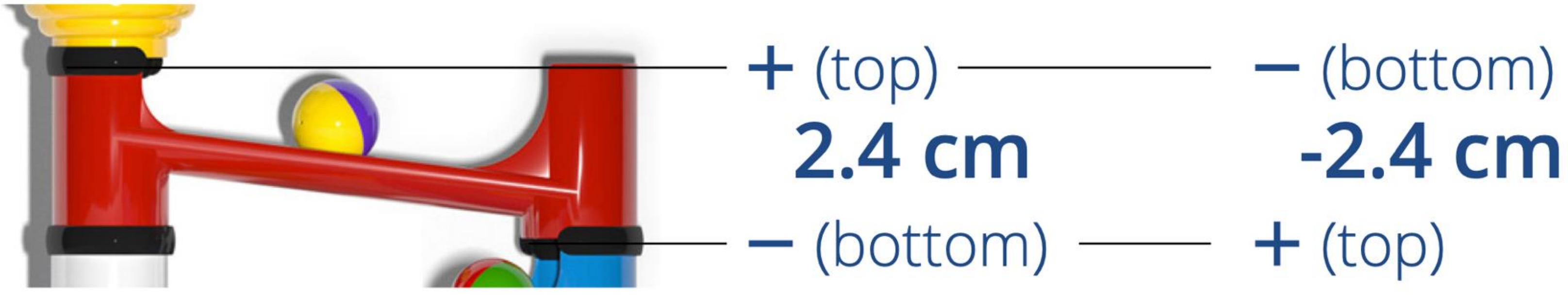
D

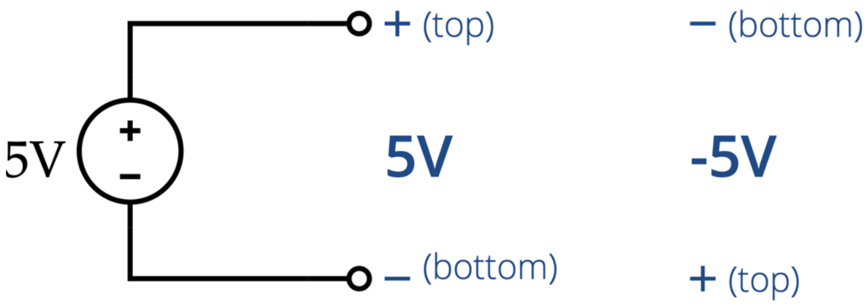


Current and voltage have signs

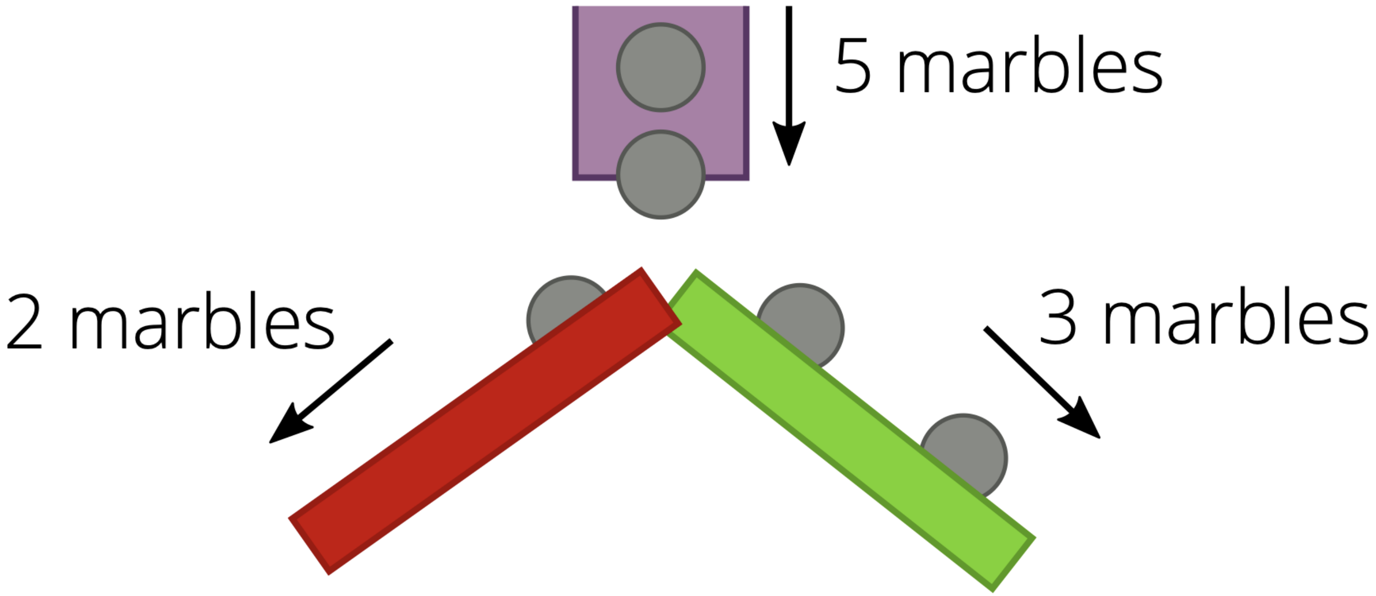


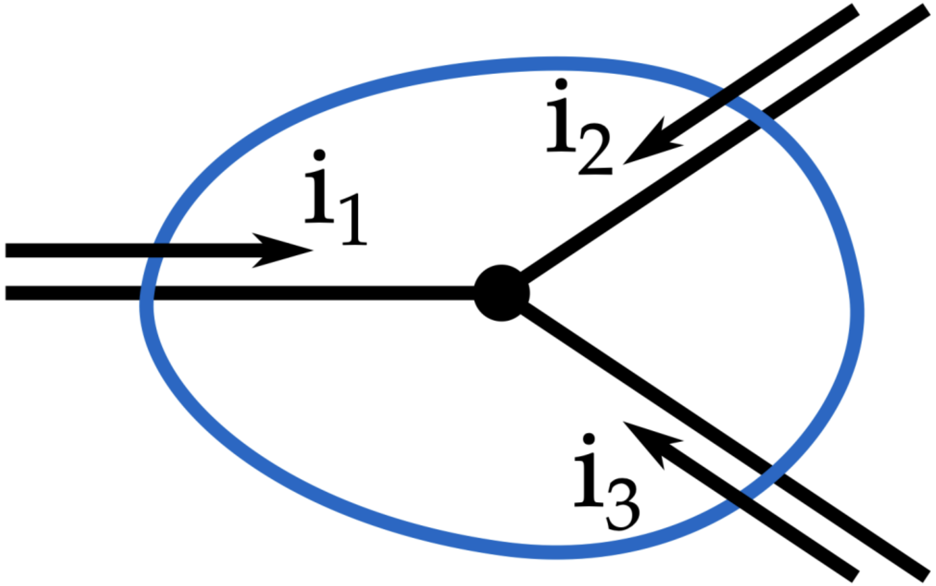


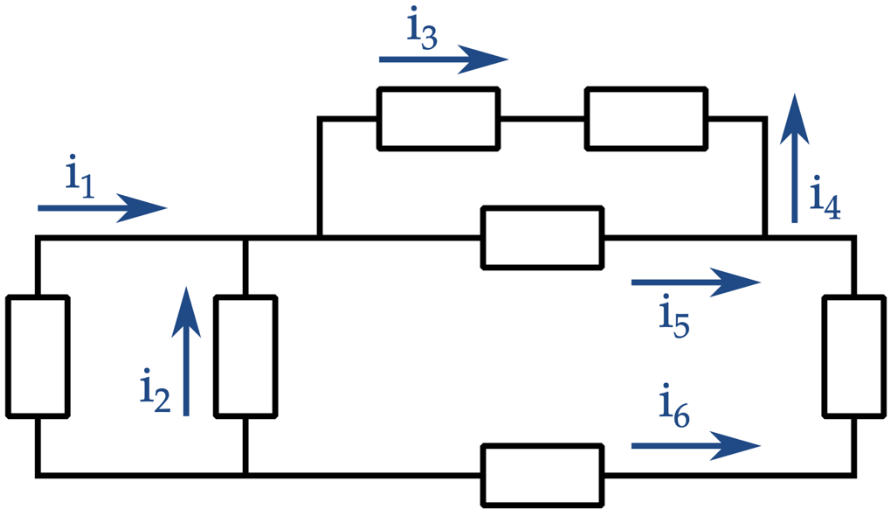




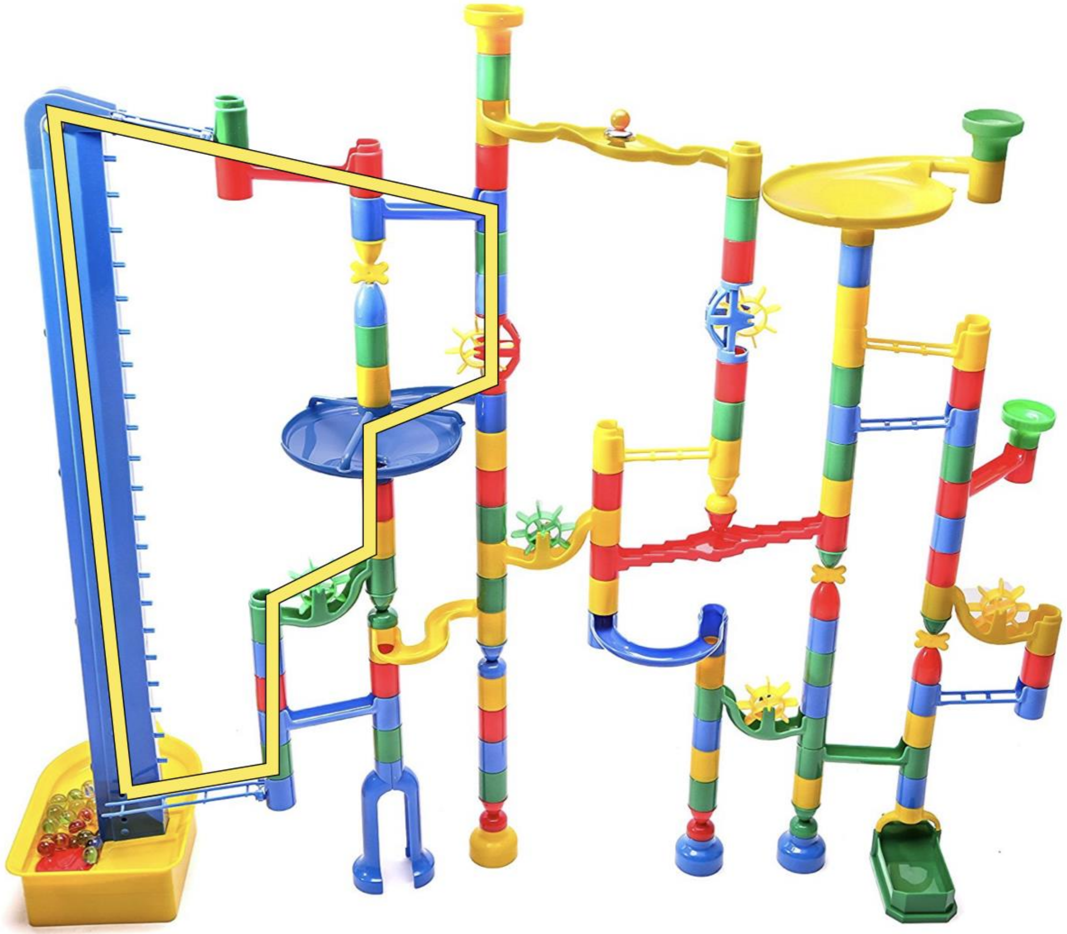
Kirchoff’s current law

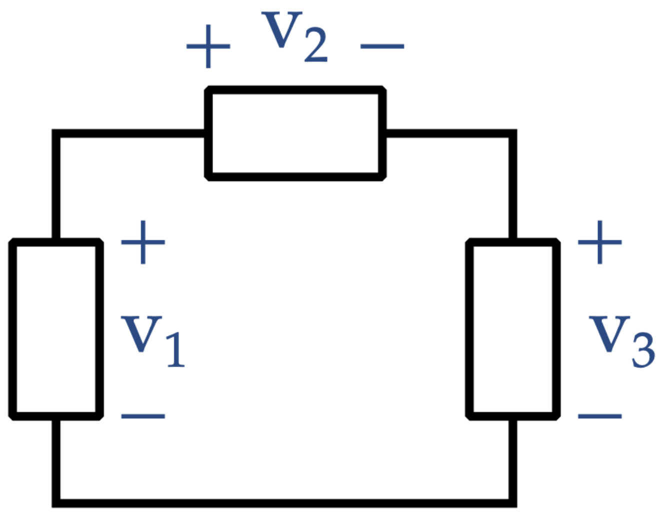


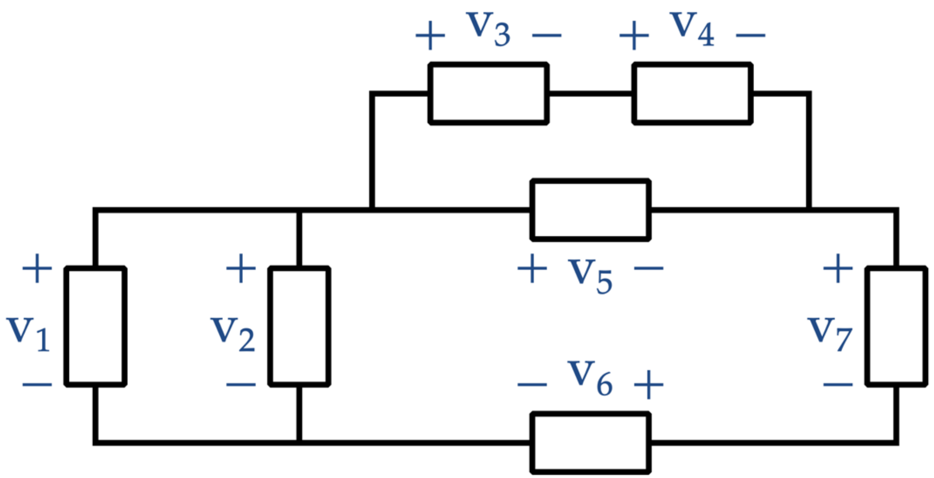




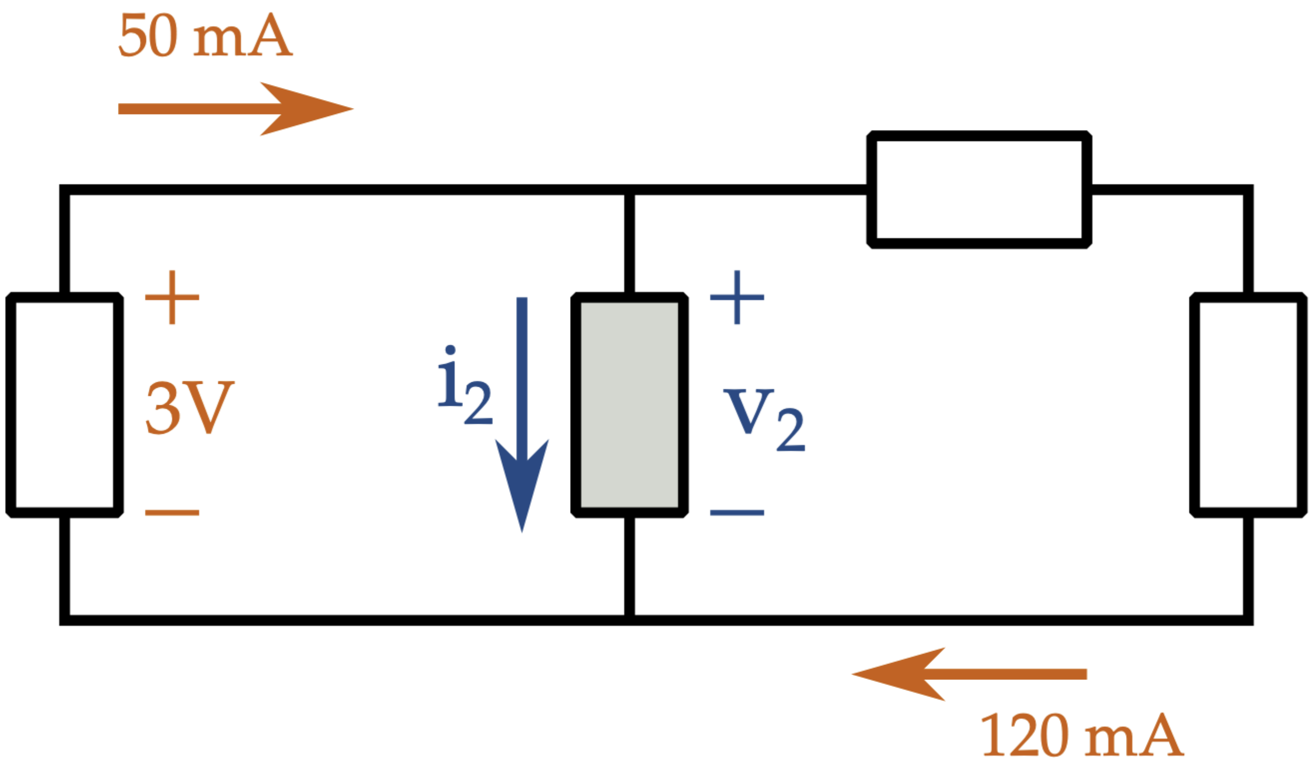
Kirchoff’s voltage law





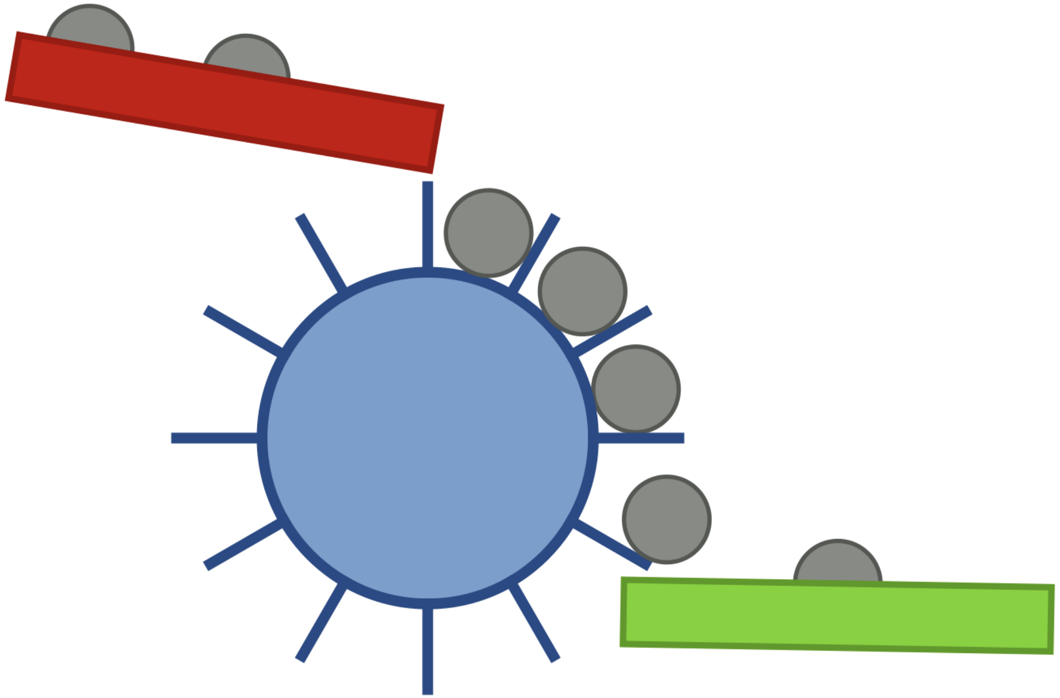


Find the voltage V2 and current I2



Is the gray element absorbing or supplying electrical energy?

Joule’s law of electric power



Power (P) = Voltage (V) × Current (I)
at the unit of Watt
(abbreviated as W)

How much power does a light bulb use,
if it is connected to 120V and draws 0.5 A?

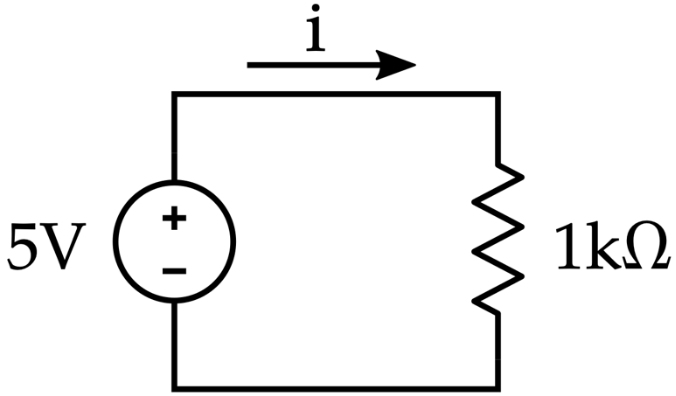
Resistor and Ohm's law

Marble-track resistor

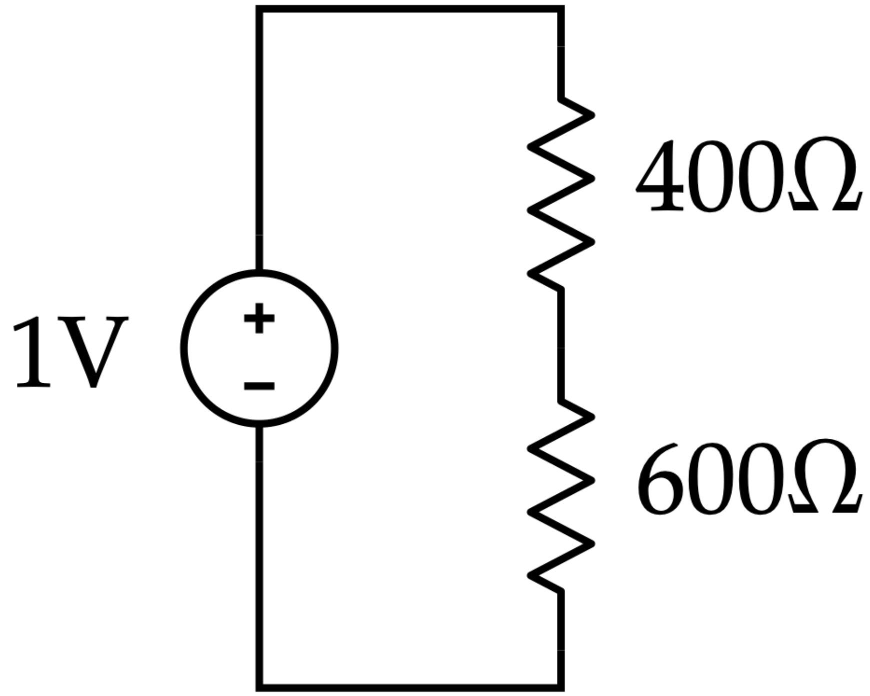
Ohm's law:

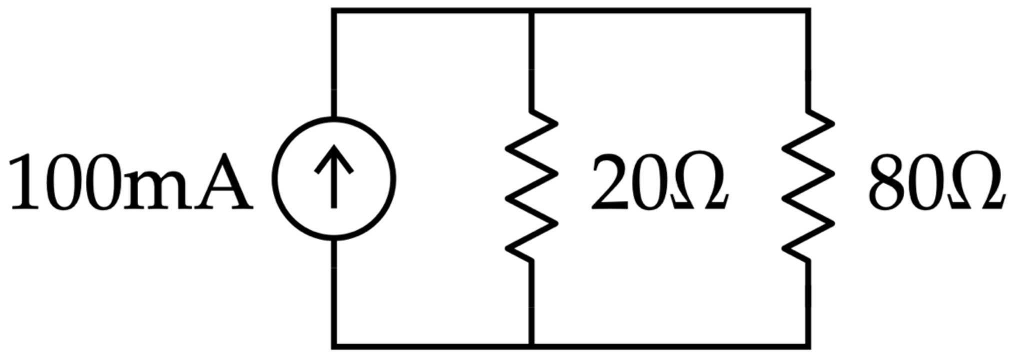
I = V / R

Current = Voltage / Resistance



Series/parallel resistors



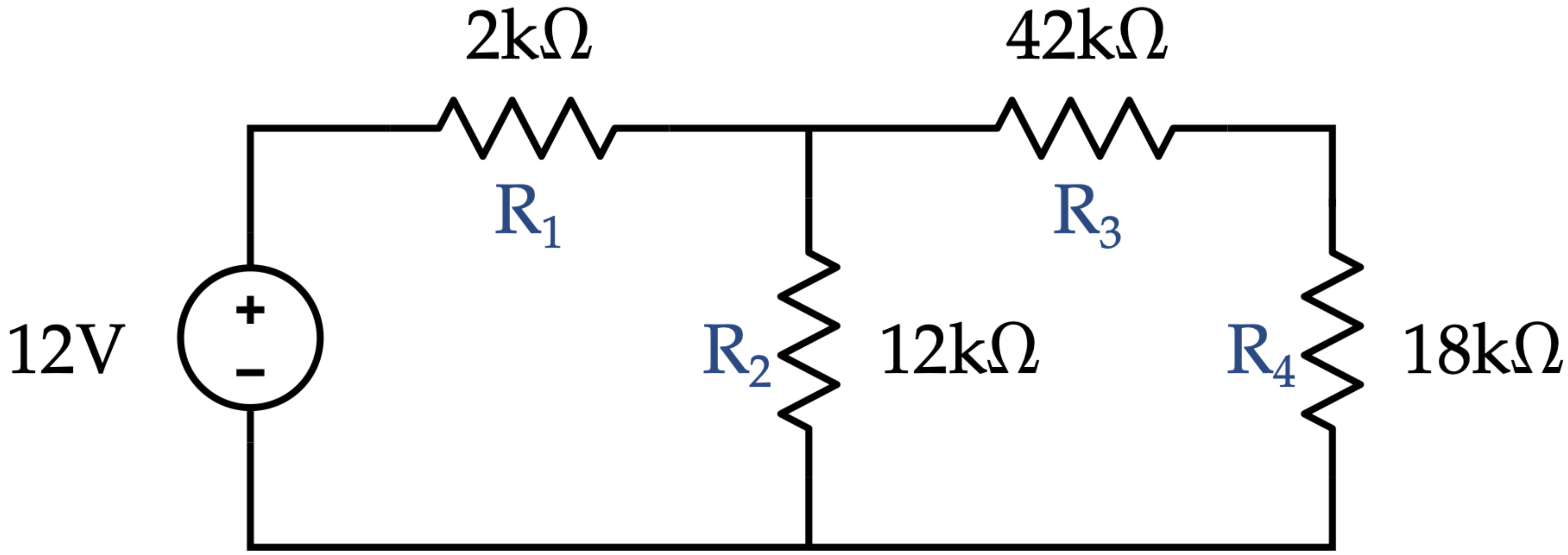


$$\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2}$$

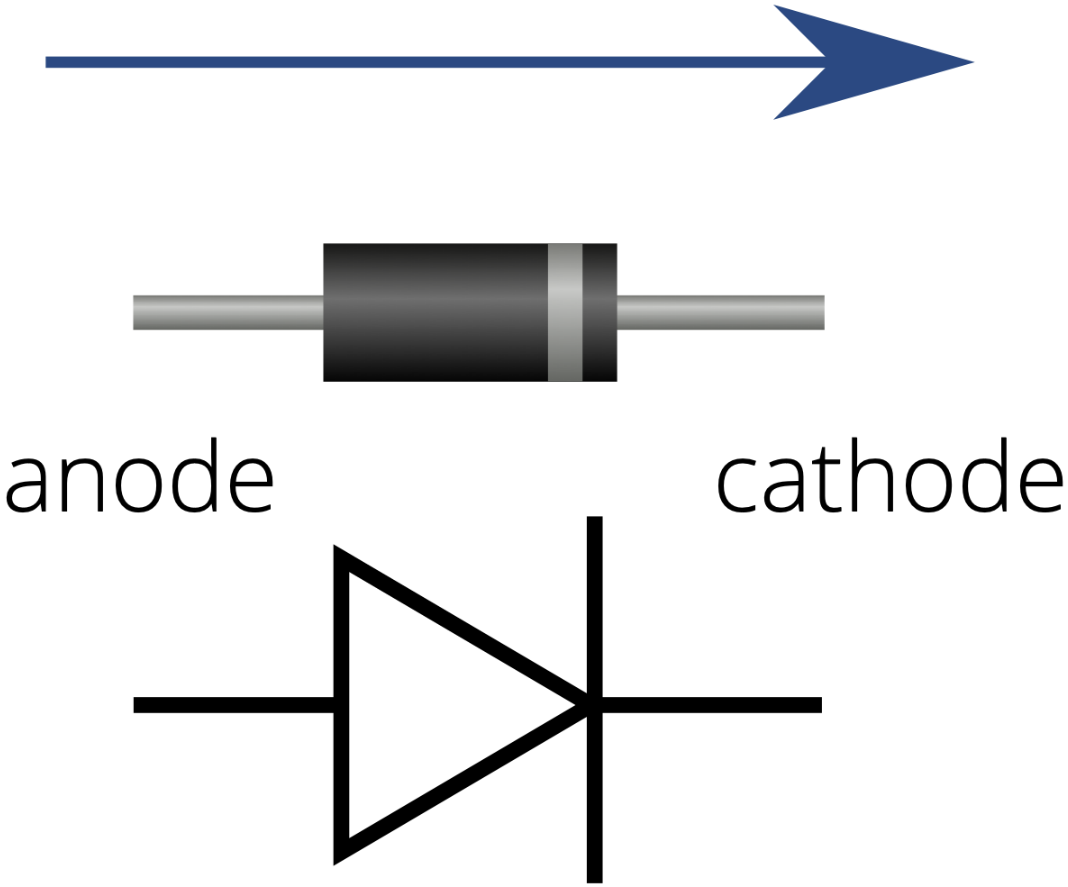
$$R_{eff} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

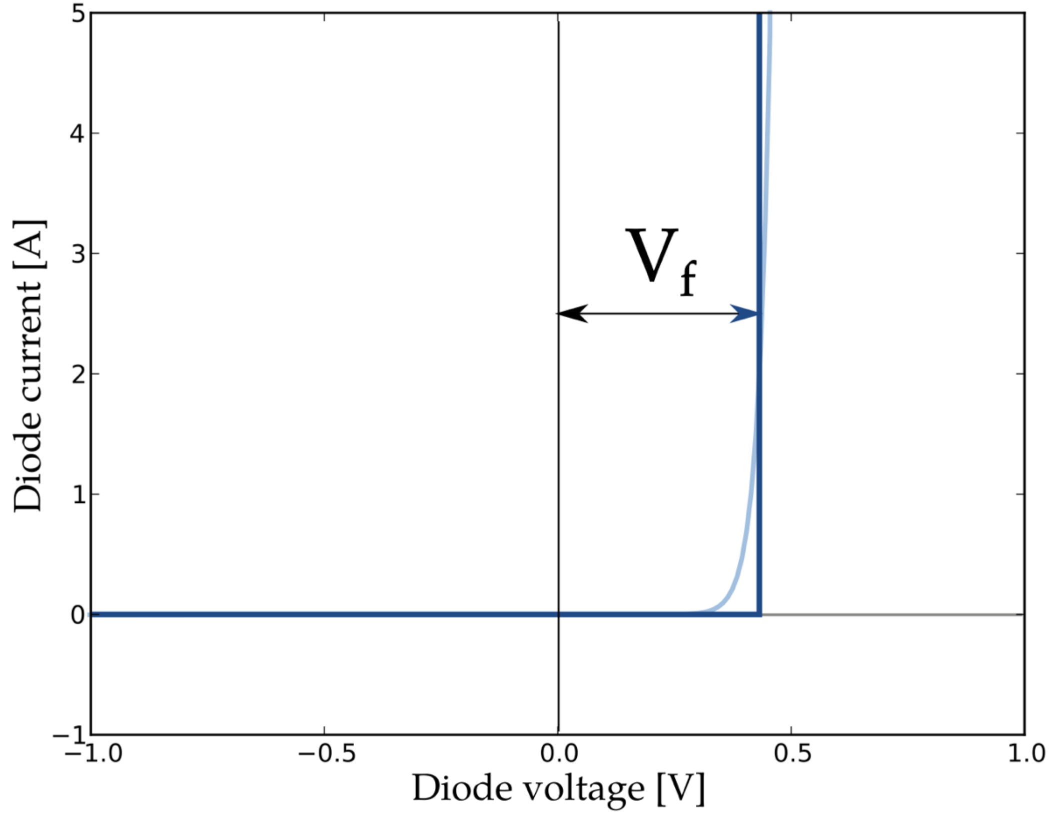
$$R_{eff} = R_1 + R_2$$

Practice

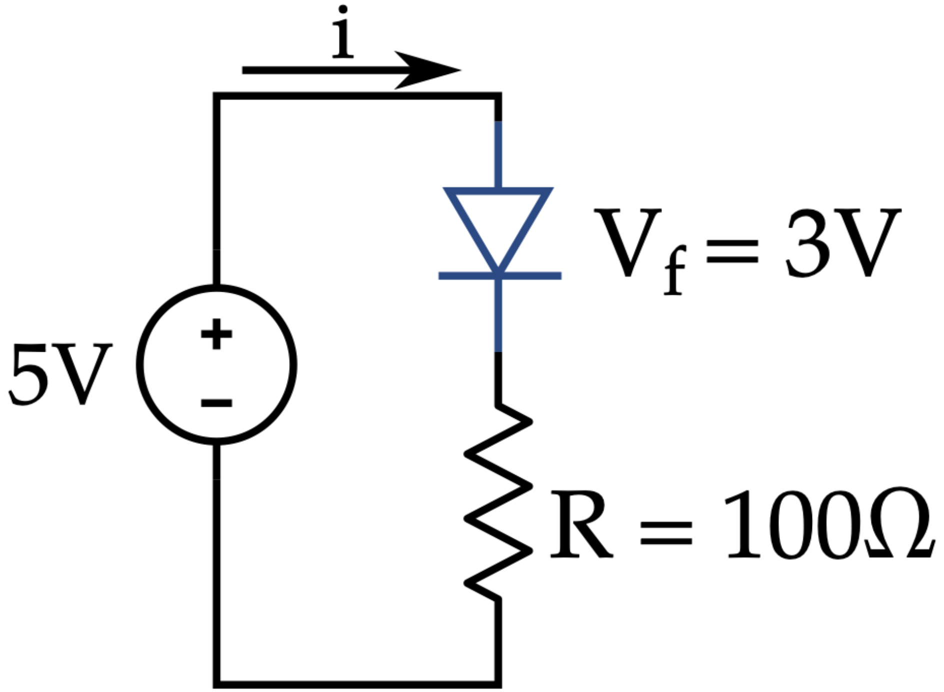


Diode: One-way valves for current





Diode practice



- What is the current through the diode?
- What is the voltage at the resistor?

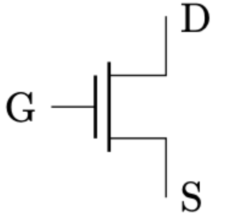
Transistor

- A transistor is an electronic device that is used to allow one electrical signal to control another electrical signal

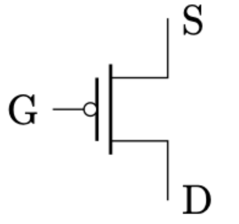
- Metal–oxide–semiconductor field-effect transistor (MOS/MOSFET)

- MOSFETs come in two types: the n-channel MOSFET (nMOS) and the p-channel MOSFET (pMOS)

- Three terminals: the gate, drain and source

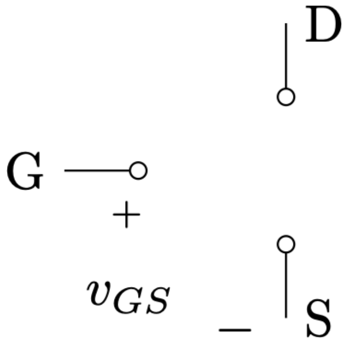


nMOS

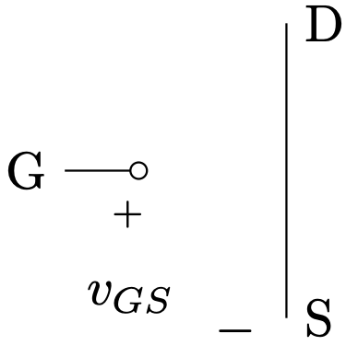


pMOS

Ideal MOS transistor switch



nMOS off: $v_{GS} < V_{th}$



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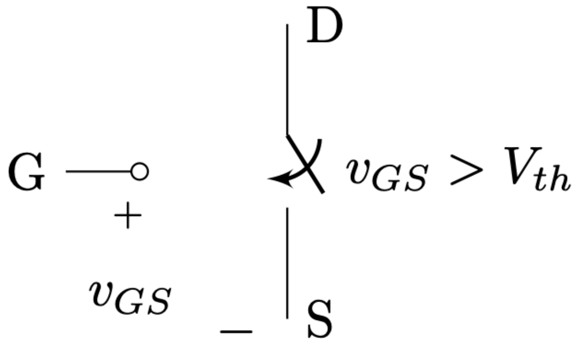
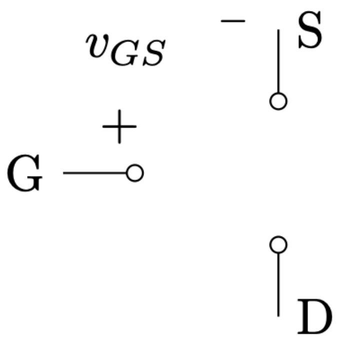
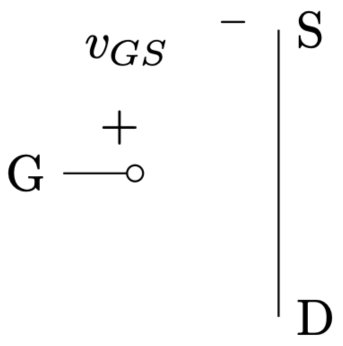


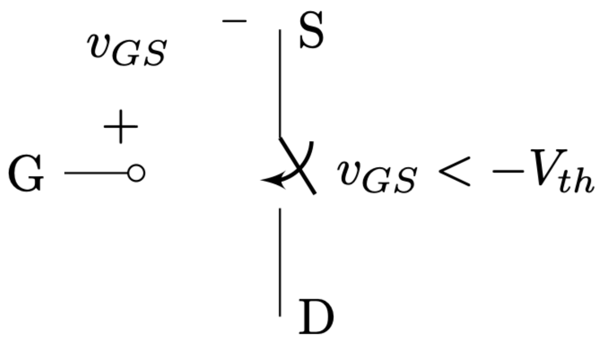
nMOS model



pMOS off: $v_{GS} > -V_{th}$

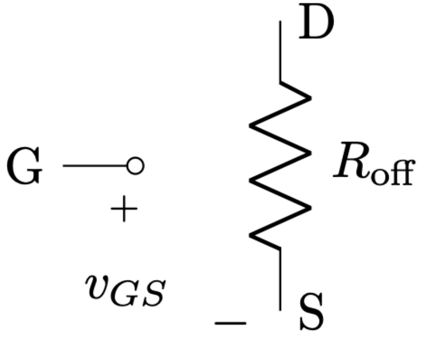


pMOS on: $v_{GS} < -V_{th}$

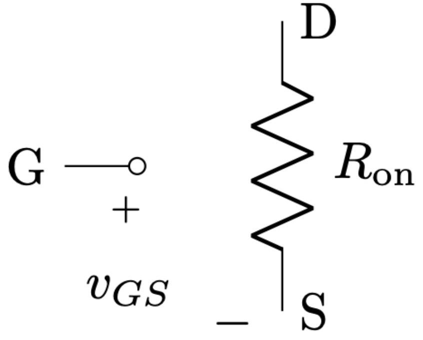


pMOS model

Modeling internal resistance



nMOS off: $v_{GS} < V_{th}$



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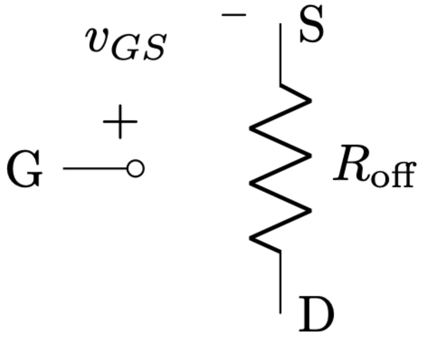
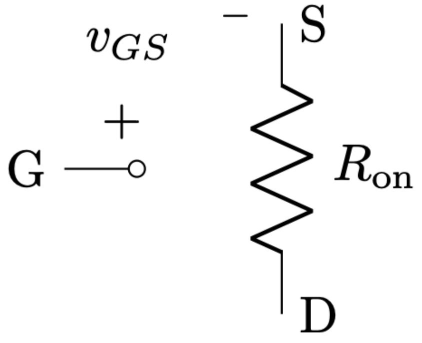


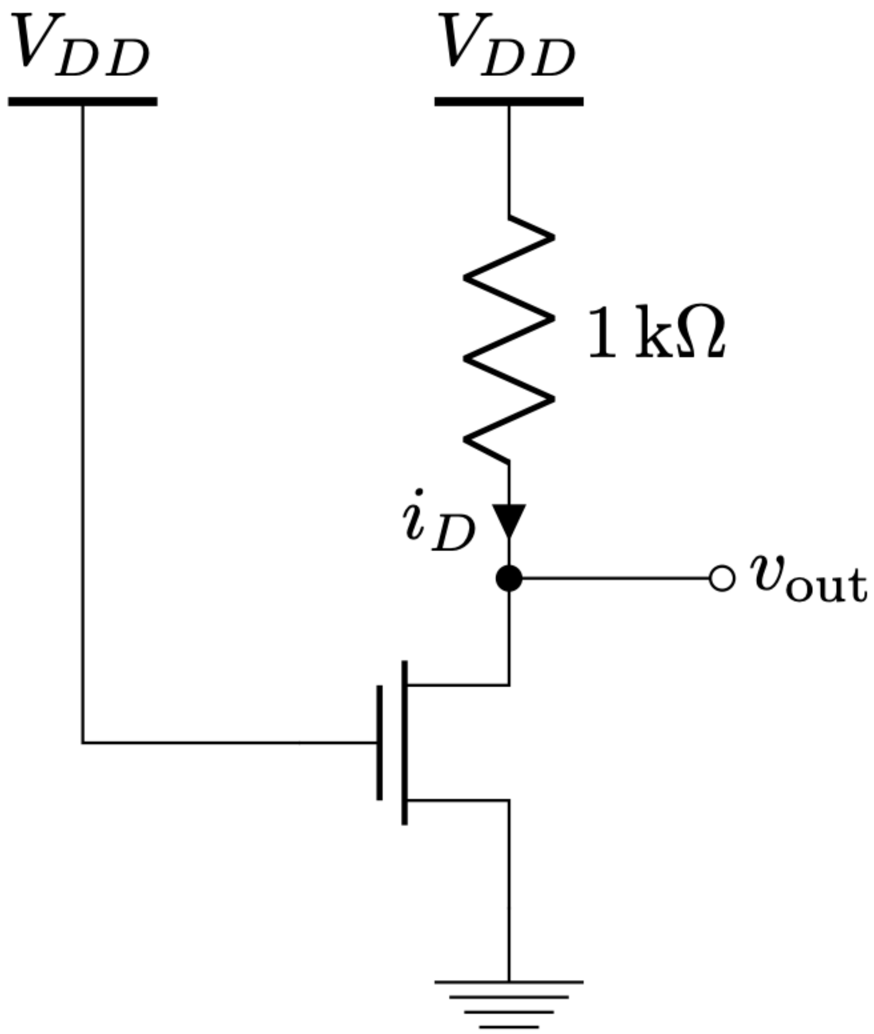
pMOS off: $v_{GS} > -V_{th}$



pMOS on: $v_{GS} < -V_{th}$

Practice

- VDD=5V, Vth=1V for transistors
- What is vout?
- What is iD?



Practice

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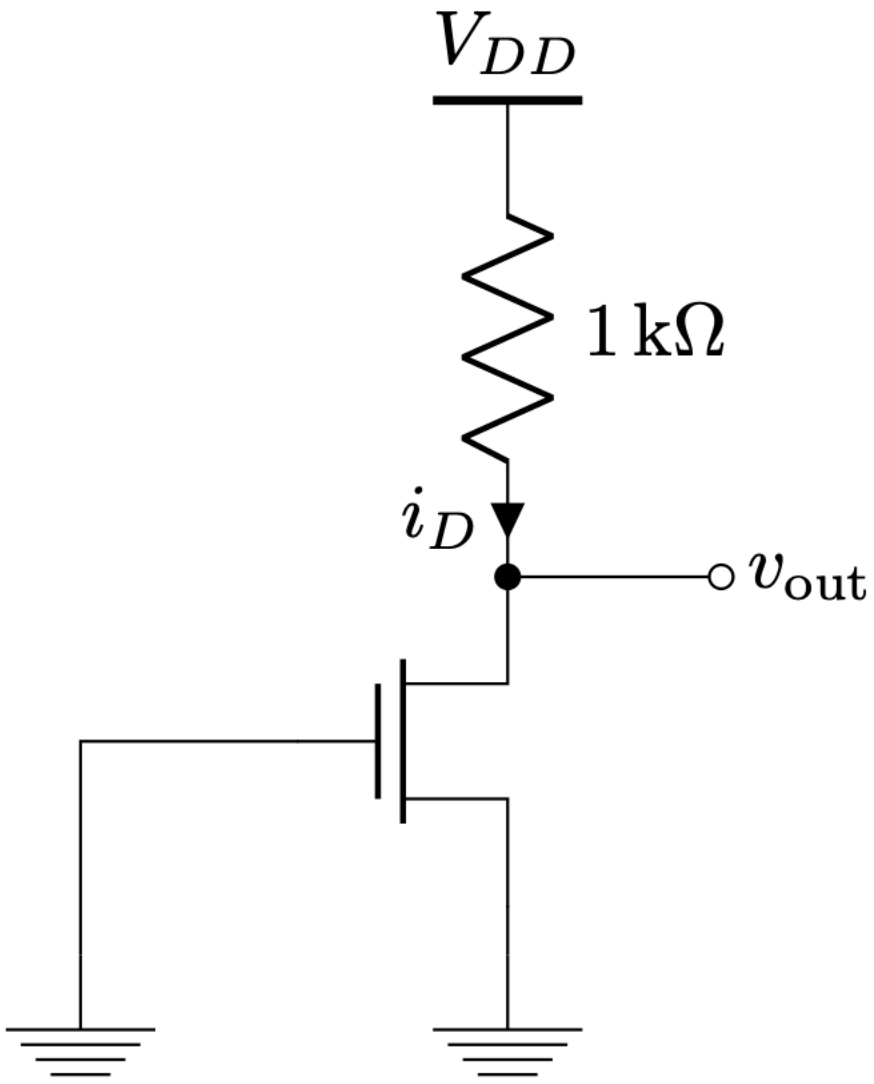
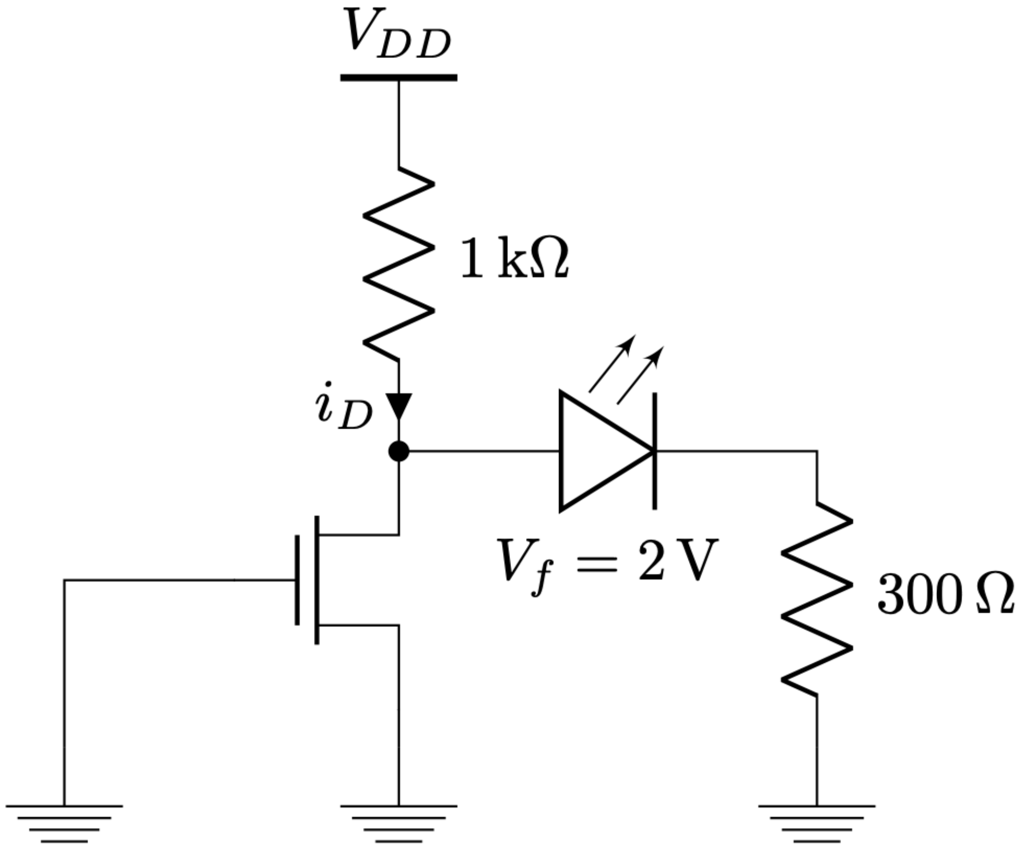


Practice

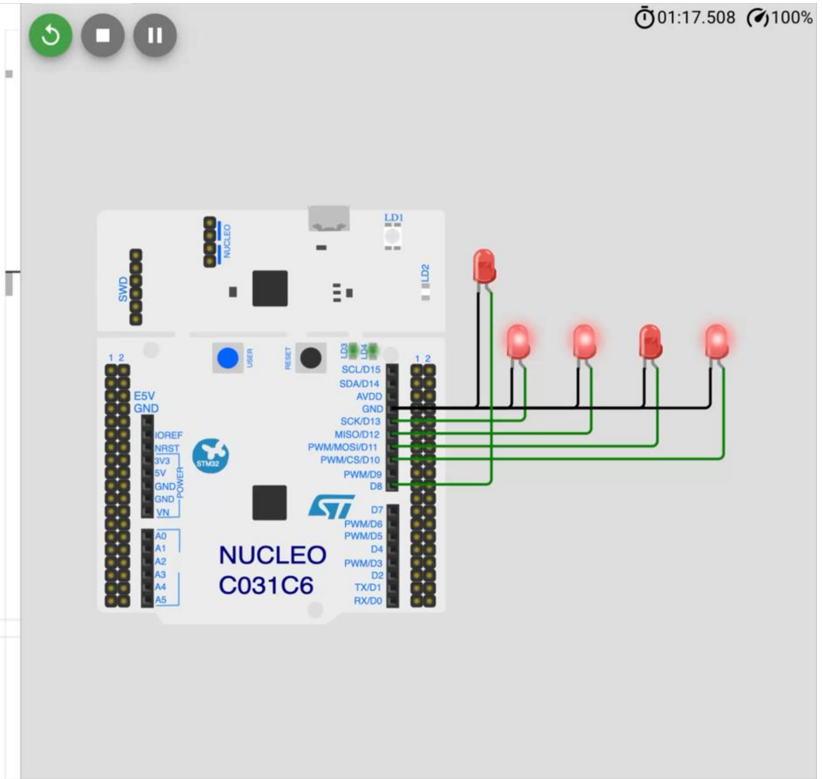
- VDD=5V, Vth=1V for transistors
- What is the current through the LED?



Recap: Running LEDs

- https://wokwi.com/projects/447390165530456065
- Can you light up one more LED only when the last two running LEDs are on?

```cpp
// 4-bit binary counter using LEDs (each LED toggles at different rates)

int ledPin[] = {13, 12, 11, 10};                // LED pins: MSB → LSB
int ledDelay[] = {8000, 4000, 2000, 1000};      // toggle interval (ms)
unsigned long lastChangeTime[4];                // store last toggle time of each LED
int ledState[4] = {LOW, LOW, LOW, LOW};          // store current state (LOW/HIGH)
unsigned long currTime;                          // current time variable

void setup() {
  currTime = 0;                                  // initialize timer
  for (int i = 0; i < 4; i++) {
    pinMode(ledPin[i], OUTPUT);                  // set pin as output
    digitalWrite(ledPin[i], ledState[i]);        // turn LED off initially
    lastChangeTime[i] = millis();                // record initial time
  }
  pinMode(8, OUTPUT);
  digitalWrite(8, LOW);
}

void loop() {
  currTime = millis();                           // get current time

  for (int i = 0; i < 4; i++) {                  // check all LEDs
    if (currTime - lastChangeTime[i] >= ledDelay[i]) {
      // time to toggle the LED
      ledState[i] = !ledState[i];                // flip LED state
      digitalWrite(ledPin[i], ledState[i]);      // apply new state
      lastChangeTime[i] = currTime;              // reset timer for that LED
    }

    if (ledState[0] == HIGH && ledState[1] == HIGH &&
        ledState[2] == HIGH && ledState[3] == HIGH) {
      digitalWrite(8, HIGH);
    } else {
      digitalWrite(8, LOW);
    }
  }
}
```



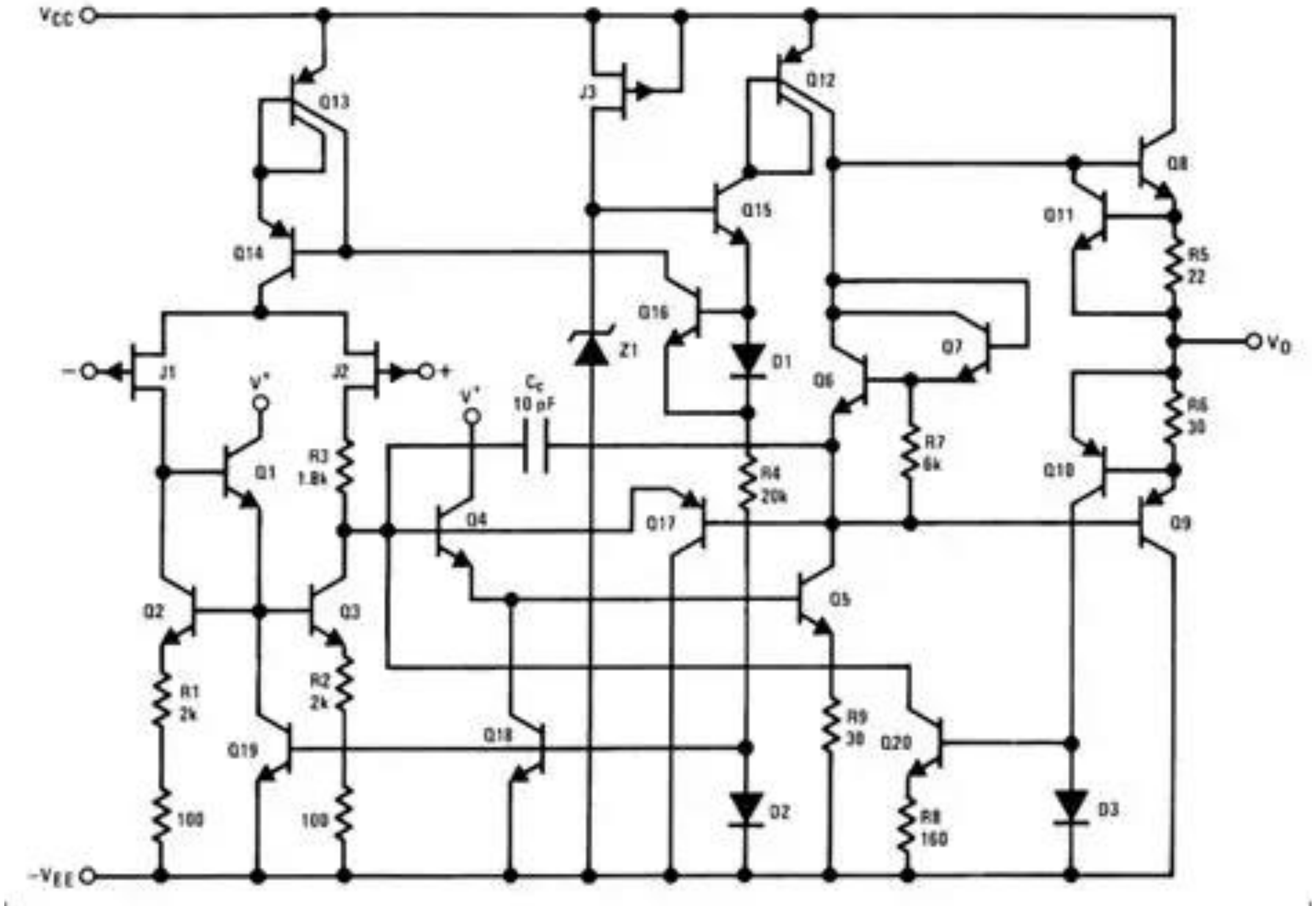
Virtual lab session: Running LEDs with digital logics

Virtual lab session: Running LEDs with digital logics

- Can you light up one more LED only when the all four running LEDs are on?

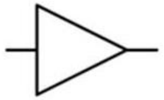
Recap: Fundamental unit: Electronic circuit (transistor)

Circuit diagram



Digital logic

Buffer

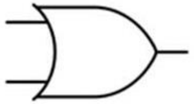


Input	Output
0	0
1	1

AND

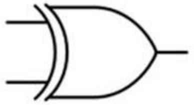
A	B	Output
0	0	0
1	0	0
0	1	0
1	1	1

OR



A	B	Output
0	0	0
1	0	1
0	1	1
1	1	1

XOR



A	B	Output
0	0	0
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Recap: Transistor

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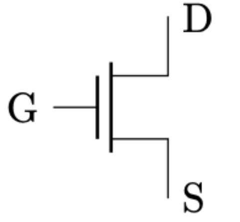
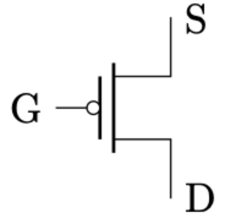
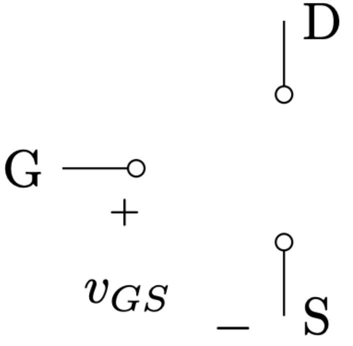


nMOS

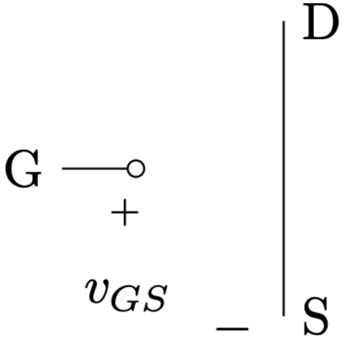


pMOS

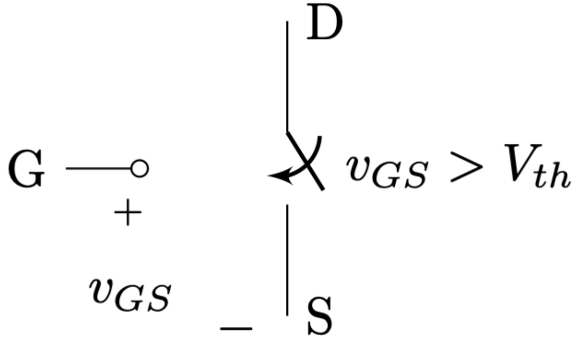
Recap: Ideal MOS transistor switch



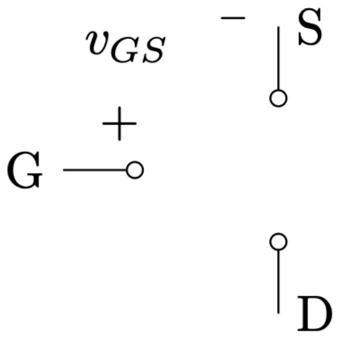
nMOS off: $v_{GS} < V_{th}$



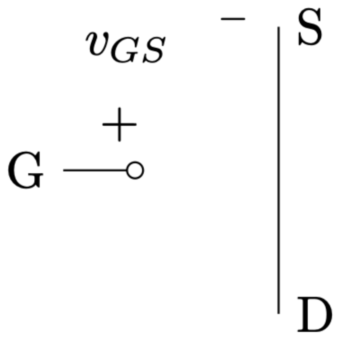
nMOS on: $v_{GS} > V_{th}$



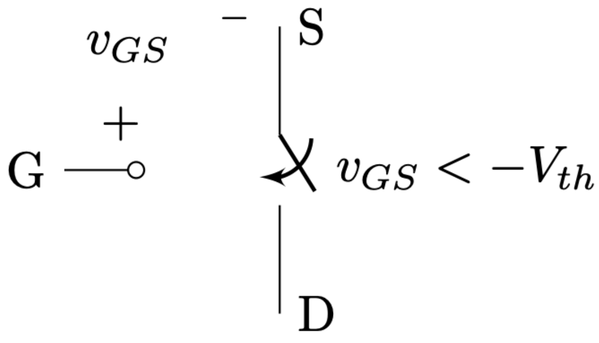
nMOS model



pMOS off: $v_{GS} > -V_{th}$



pMOS on: $v_{GS} < -V_{th}$



pMOS model

Information is stored on the computer in binary form

- An information variable represented by physical quantity
- For digital systems, the variable takes on discrete values
- Two level, or binary values are the most prevalent values in digital systems
- Binary values are represented by values or ranges of values of physical quantities

Binary logic Implemented with MOS

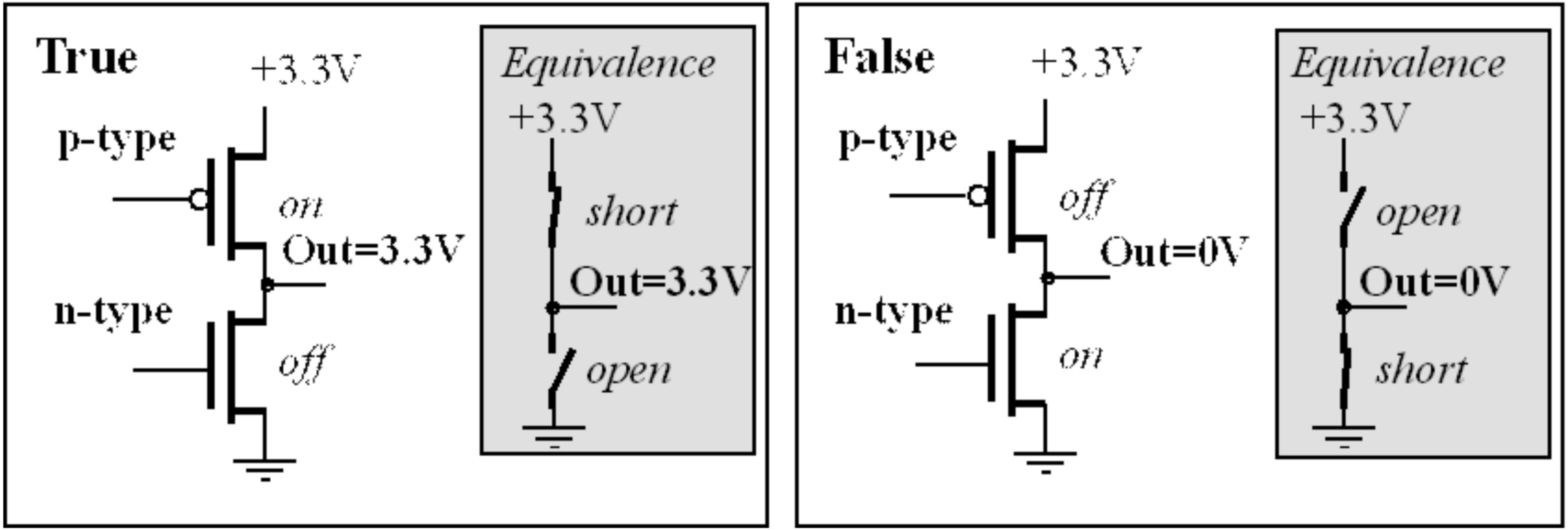
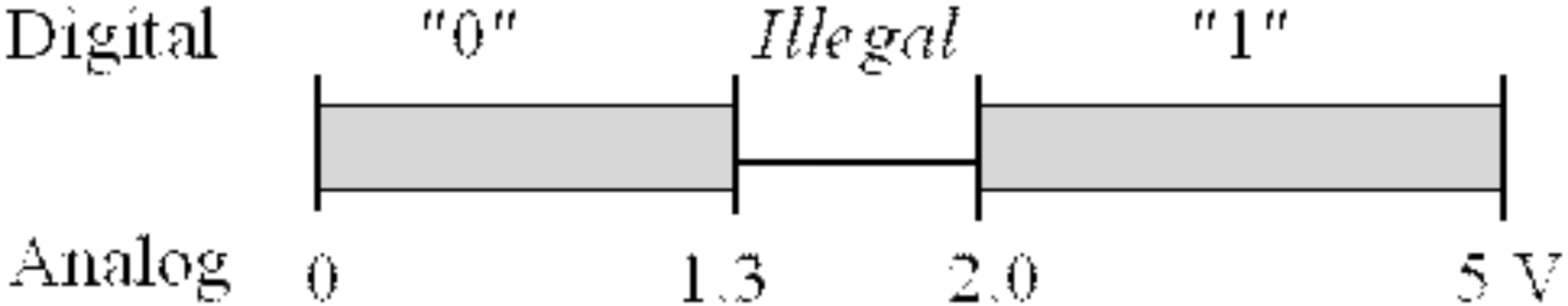
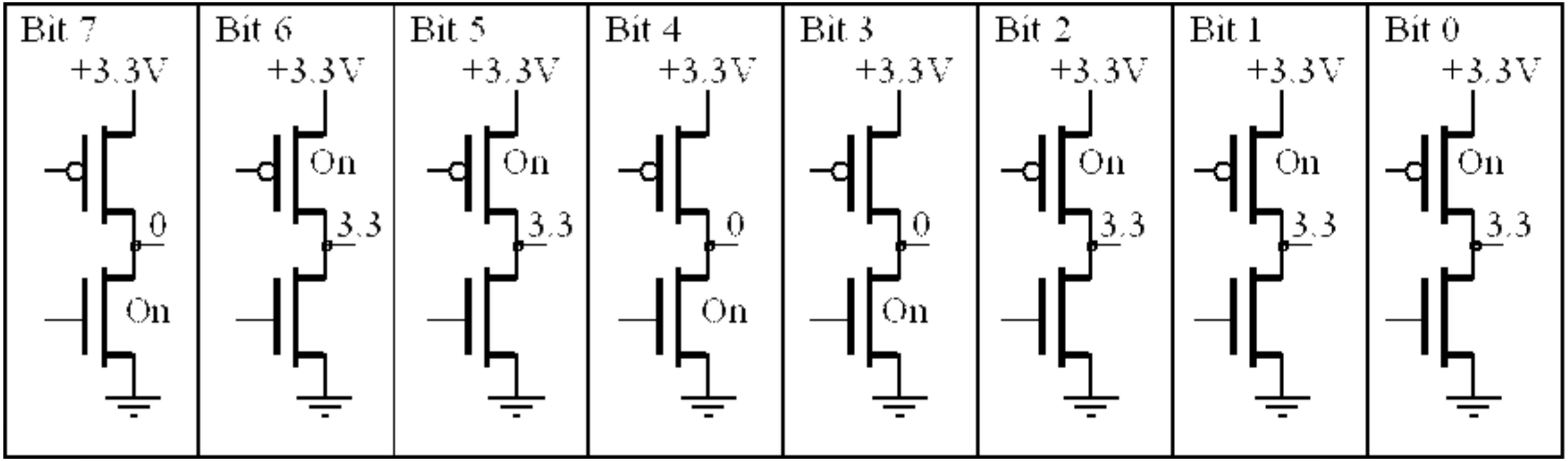


Figure 4.1. A binary bit is true if a voltage is present and false if the voltage is 0.

Binary logics are represented by values or ranges of values of physical quantities



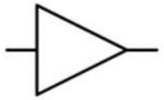
An implementation of 01100111



Digital logic of NOT gate

Digital logic

Buffer

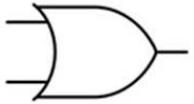


Input	Output
0	0
1	1

AND

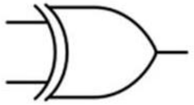
A	B	Output
0	0	0
1	0	0
0	1	0
1	1	1

OR



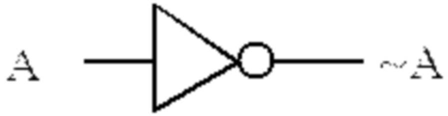
A	B	Output
0	0	0
1	0	1
0	1	1
1	1	1

XOR

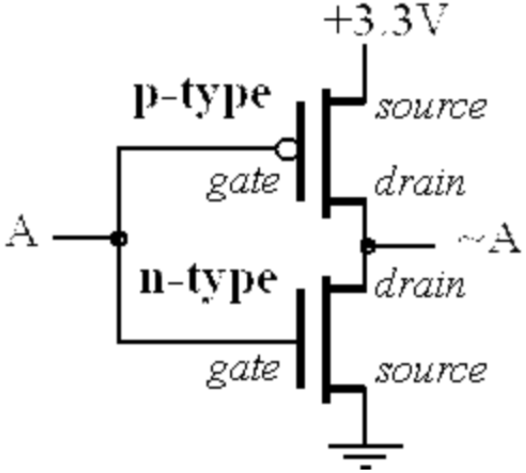


A	B	Output
0	0	0
1	0	1
0	1	1
1	1	0

NOT gate

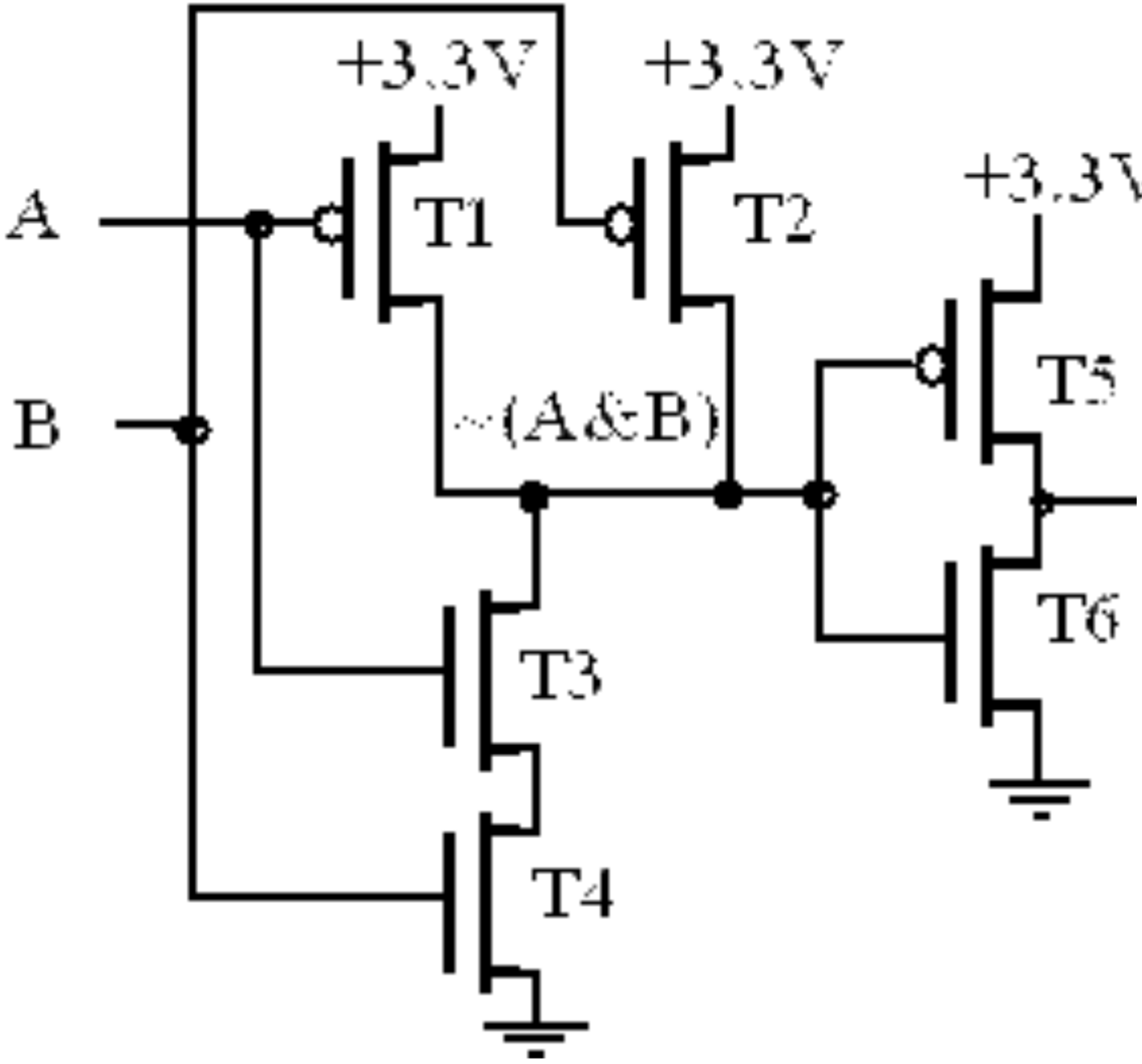


A	~A
0	1
1	0

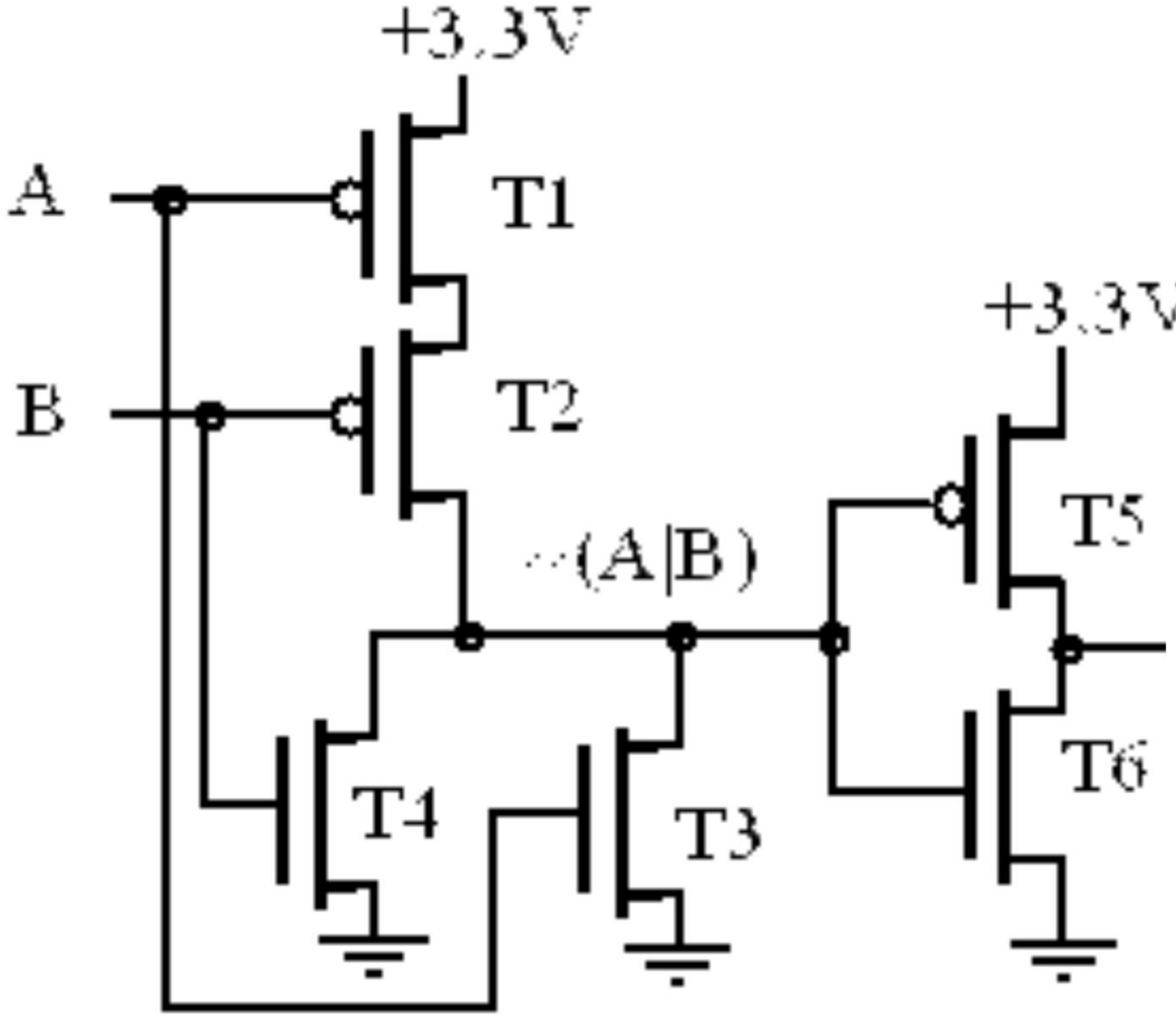


A	p-type	n-type	~A
0 V	active	off	+3.3V
+3.3V	off	active	0V

Practice



Practice



Digital logics by bit or Boolean

$$\begin{array}{lr} A = & 01100111 \\ B = & \underline{11110000} \\ A \& B & 01100000 \end{array}$$

$$\begin{array}{lr} A = & 01100111 \\ B = & \underline{11110000} \\ A \mid B & 11110111 \end{array}$$

$$\begin{array}{lr} A = & 01100111 \\ B = & \underline{11110000} \\ A \&\& B & 1 \end{array}$$

$$\begin{array}{lr} A = & 01100111 \\ B = & \underline{11110000} \\ A \mid\mid B & 1 \end{array}$$

Boolean Algebra

A & B = B & A	Commutative Law
A | B = B | A	Commutative Law
(A & B) & C = A & (B & C)	Associative Law
(A | B) | C = A | (B | C)	Associative Law
(A | B) & C = (A & C) | (B & C)	Distributive Law
(A & B) | C = (A | C) & (B | C)	Distributive Law
A & 0 = 0	Identity of 0
A | 0 = A	Identity of 0
A & 1 = A	Identity of 1
A | 1 = 1	Identity of 1
A | A = A	Property of OR
A | (~A) = 1	Property of OR
A & A = A	Property of AND
A & (~A) = 0	Property of AND
~(~A) = A	Inverse
~(A | B) = (~A) & (~B)	De Morgan's Theorem
~(A & B) = (~A) | (~B)	De Morgan's Theorem

Practice

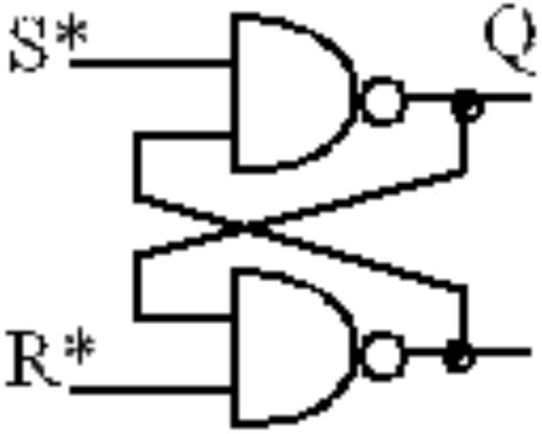
A	B	C	A & (B | C)
$0100_2 = 4$	$0101_2 = 5$	$0110_2 = 6$	
$0111_2 = 7$	$1010_2 = 10$	$0001_2 = 1$	
$1110_2 = 14$	$1001_2 = 9$	$0111_2 = 7$	

Practice

A	B	C	A && (B || C)
True	False	False	
True	False	True	
False	True	False	
True	True	False	

Digital logic for storing a bit

Set-Reset Latch

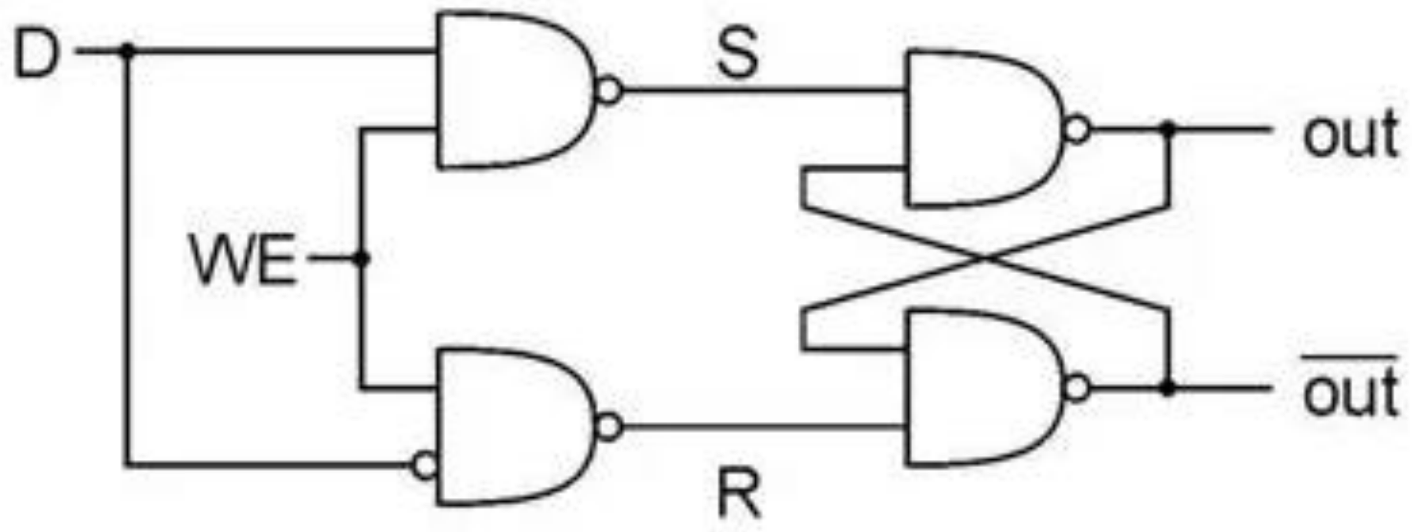


Truth Table		
Input		Output
Set	Reset	Q
Low	Low	Last State
Low	High	Low
High	Low	High
High	High	Unpredictable

Digital logic for storing a bit

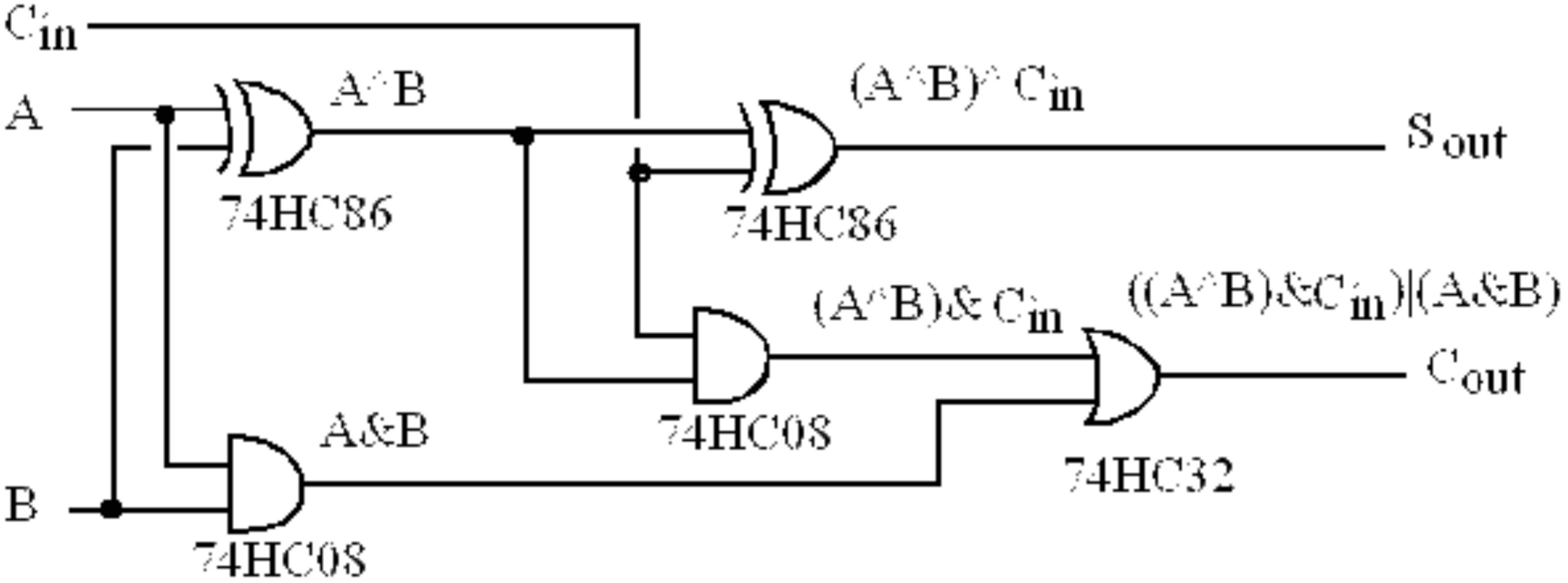
Two inputs: D (data) and WE (write enable)

- **when WE = 1, latch is set to value of D**
  - ➢ **S = NOT(D), R = D**
- **when WE = 0, latch holds previous value**
  - ➢ **S = R = 1**

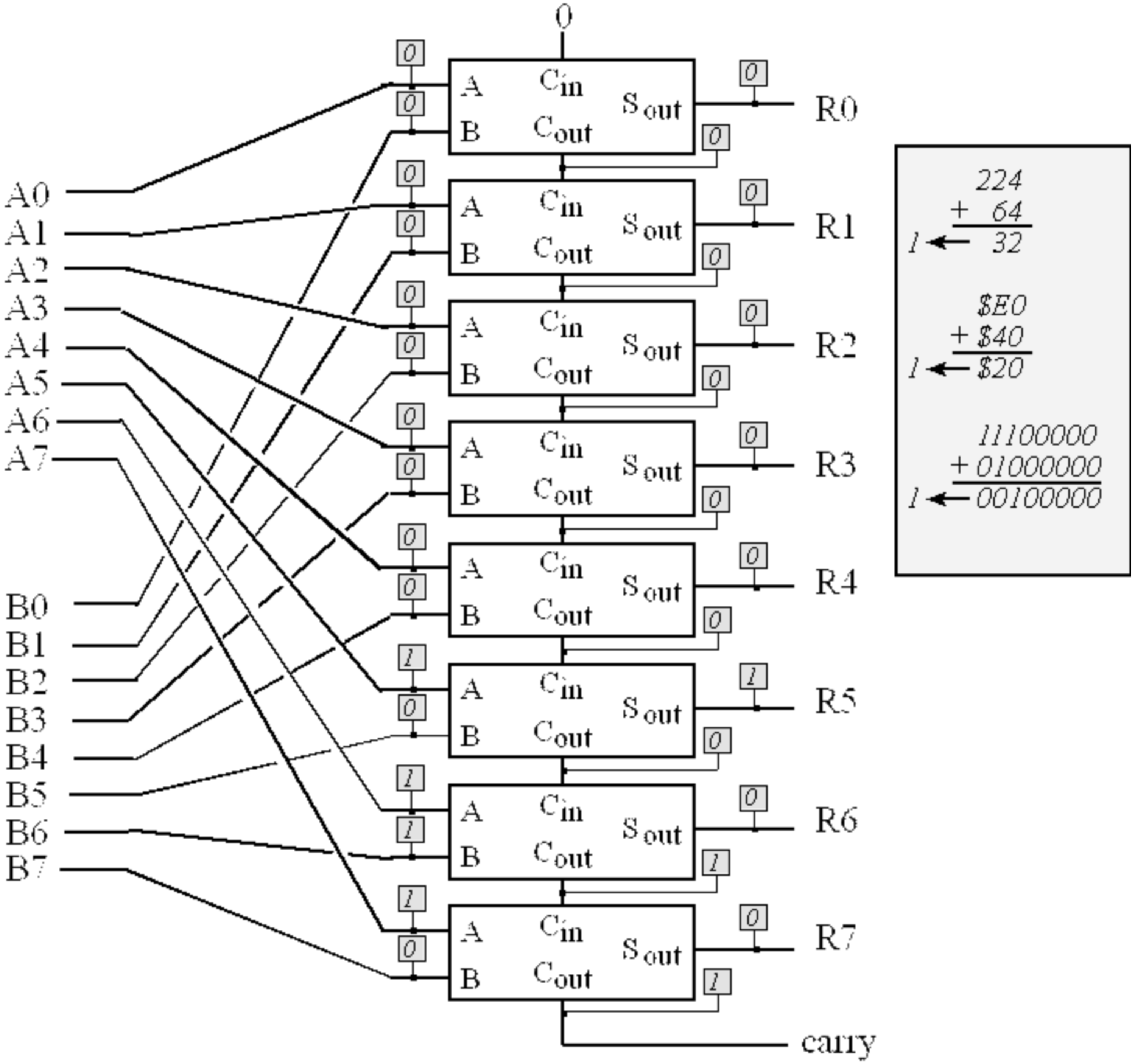


Digital information stored in memory

Binary adder



8-bit adder


0
A0
A1
A2
A3
A4
A5
A6
A7
B0
B1
B2
B3
B4
B5
B6
B7
A
B
Cin
Cout
Sout
R0
R1
R2
R3
R4
R5
R6
R7
carry
224
+ 64
1 ← 32
$E0
+ $40
1 ← $20
11100000
+ 01000000
1 ← 00100000

Common abbreviations for large numbers

Value	*SI Decimal*	*SI Decimal*
1000^1	k	kilo-
1000^2	M	mega-
1000^3	G	giga-
1000^4	T	tera-
1000^5	P	peta-
1000^6	E	exa-
1000^7	Z	zetta-
1000^8	Y	yotta-

Value	*IEC Binary*	*IEC Binary*
1024^1	Ki	kibi-
1024^2	Mi	mebi-
1024^3	Gi	gibi-
1024^4	Ti	tebi-
1024^5	Pi	pebi-
1024^6	Ei	exbi-
1024^7	Zi	zebi-
1024^8	Yi	yobi-

Recap: Running LEDs with digital logics

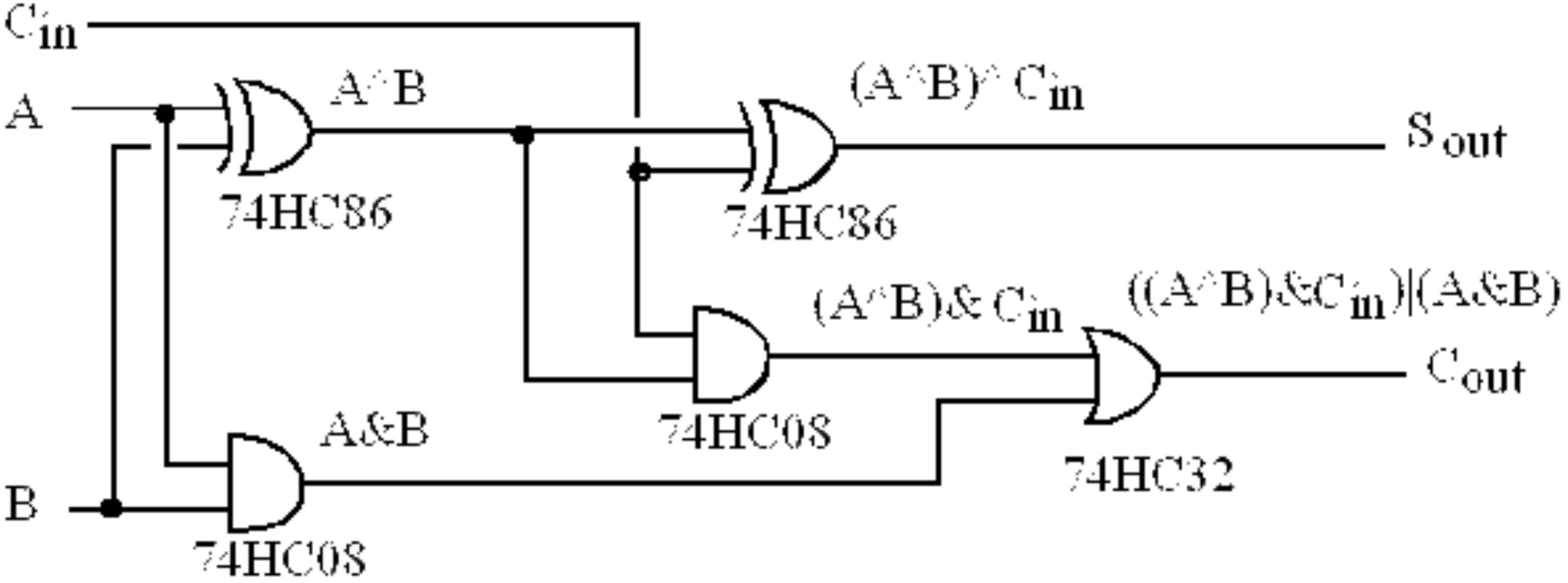
- https://wokwi.com/projects/447390165530456065

Virtual lab session: Binary adder for LED1 and LED2

- Can you implement a binary adder for LED1 and LED2?

Virtual lab session: Binary adder for LED1 and LED2

- Can you implement a binary adder for LED1, LED2 and LED3 (carry) ?



Thank you!